

MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATICS



Bachelor of Science in Software Engineering [BSc. SE] Degree Program (Full Time)

AUGUST 2018

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SECTION A

1. Introduction and Background

Mbarara University of Science and Technology (MUST) is located within Mbarara Municipality 267 km from Kampala along Kampala-Kabale highway, in Mbarara District-Southwestern Uganda. Mbarara University of Science and Technology, the second public university founded in Uganda, is accredited by the country’s National Council for Higher Education (NCHE). Since the inception of Mbarara University of Science and Technology, a good level of infrastructural growth and program expansion has been registered. The university is currently running Certificate, Diploma, Undergraduate, Post Graduate Diploma, Masters and PhD research programs. The university vision is “To provide quality and relevant education at national and international level with particular emphasis on Science and Technology, and its application to community development”. Since its establishment in 1989, MUST has grown to a multi-disciplinary university with more than 500 academic/research staff and 3,000 students.

At present, MUST has two major campuses; the main campus (current location of all Faculties and Institutes) and Kihumuro campus (planned Campus for MUST). There are two academic faculties and four institutes namely Faculty of Medicine, Faculty of Science, Faculty of computing and informatics sciences, Institute of Inter-disciplinary Training and Research, and the Institute of Management Sciences. The University has the following well established satellite training centers:

1. The Institute of Tropical Forest Conservation (ITFC) in Bwindi Impenetrable National forest for educational purposes and research in areas of tropical forest conservation and herbal medicine under the Faculty of Science.
2. The Rugazi and Bughoye Teaching Health Centers under Faculty of Medicine are community based centers for training third and fourth year Medical students.

1.1 Vision and mission of MUST

Our vision is: To become a center for academic and professional excellence in science and technology

Mission:

To provide quality relevant education at national and international level with particular emphasis on science and technology and its application to community development.

1.2 Faculty of Computing and Informatics Sciences

The Faculty of computing and informatics sciences was established in 1997 under the Faculty of Science and later shifted to the Faculty of Development Studies to run courses in Computer Science to meet the increasing demand of ICT in the country. The Faculty started with the Department of Computer Science running a Bachelor of Computer Science Course. In September 2006, the Department of Computer Science was then elevated to the status of the Institute of Computer Science. The Faculty aspires to be a university's great Computer Science and Information Technology centre recognized nationally and internationally for excellence in teaching community outreach programs with multi-disciplinary dimensions and research

approach. The Institute now offers majorly undergraduate degree programs and technical short courses to meet the demand for Computer and IT basic skills. The programs offered include:

1. Undergraduate Degree Programs: Bachelor of Information Technology (BIT), Bachelor of Science in Information Technology (B) and Bachelor of Science in Computer Engineering and now, bachelor of software engineering.
2. Short Technical Courses: Certificate in Computer Applications (CCA), Cisco Certified Network Associate (CCNA), Cisco Certified Network Professional (CCNP), Cisco Networking Academy Program (CNAP), Information Technology Essentials (ITE) I & II and International Computer Driving License (ICDL) Certification.
3. We also have postgraduate programs among which we have Masters of Science in information systems and Master of Science in Health information Technology, all running for two years.

2.0 Program name and duration

2.1 Program name:

This is a Bachelor of Science in software engineering degree, a full time day program that will run for four years. Bachelor of software engineering has a close relationship with the rest of the undergraduate programs(BIT,BCE,BCS) in that students who do this programs will always interface with the rest though different course units that cut across these programs, among which we have computer programming, networking, data communication, project management, to mention but a few. Research in the Faculty of computing and informatics sciences has gone to a level where students from different programs can do research projects together, so as to enhance integration of knowledge and team work, and therefore, software engineering is not an exception.

2.2 Program duration:

This program is full time and is scheduled to run for four years, making up a total of 8 semesters of classwork and field work

3.0 Program description

Software Engineering is the systematic development and application of techniques which lead to the creation of correct and reliable computer software. To achieve that, you need a wide understanding of the general principles which underpin not only computer software but also computer hardware and computer communications. As our daily lives become more dependent on computer systems, it is vital that such systems are error-free and totally reliable. This reliability is particularly important when computers are used in safety-critical situations such as hospitals, or controlling aircraft or industrial plants. Software Engineering focuses on challenging problems in industry and commerce concerned with software development and reliability. Concern for software reliability, correctness, safety and cost, are the hallmark of Software Engineering. The principles and knowledge of computer science, engineering, and mathematical analysis are employed by computer software engineers for designing, developing, testing, and evaluating the software and the systems that computers use to carry out various applications.

Software engineers are engaged in analysing user needs and designing, constructing, testing, and maintaining computer applications software or systems. Various kinds of software like software for operating systems and network distribution, and compilers, which convert programs for execution on a computer, are developed by a software engineer. In the programming or coding fields, software engineers give instructions to a computer, line by line; on how to perform a function or operation. These engineers are also geared to tackle technical problems and hitches. Although these engineers need to possess strong programming skills, they are more occupied with the development of algorithms and in analysing and solving problems in programming than with writing codes. Computer software engineers are usually a part of the team that designs and develops advanced hardware, software, and systems. Thus, until a finished product is developed and released, workers from various branches including those of engineering, marketing, production and design collaborate with each other, of which software engineers are a basic part.

3.1 Career opportunities

Students may pursue career opportunities in a variety of entry level positions, including software developer, software programmer, application developer, Web application developer, software

engineer, software applications analyst, Web applications analyst, Web applications specialist and software applications specialist. The Bachelor of Software Engineering program is offered as a day program at MUST.

4.0 Program Rationale

Software engineering comprises the core principles consistent in software construction and maintenance: fundamental software processes and life-cycles, mathematical foundations of software engineering, requirements analysis and validation, software engineering methodologies and standard notations, principles of software architecture and re-use, software quality frameworks and verification, and environments and tools for the software development and maintenance.

Current industry-strength programming languages, technologies and systems feature highly in the practical components, electives and projects of the course, but they are also taught with a view to understanding and applying principles underlying their more ephemeral character.

The Bachelor of Science in Software Engineering Degree Program will be offered to give an opportunity to prospective students to undertake training in Software engineering at a bachelors' degree level within Uganda. With the growth of the ICT sector in Uganda, there is need to produce graduates who are job creators using the skills obtained in software Engineering

4.1 Needs assessment

Our Source of information for needs assessment was obtained from: Managers of Industrial training places of our students where we went consulted and learnt about a skills gap in design and implementation of good software programs.

Business owners also had inputs. Commercial businesses have the need to have softwares of good quality to sustain their businesses over a long period of time. They need personnel who can take care of the changing requirements of businesses over time. So they need adaptive software.

From Industry and production both the new entrants and those existing eg coca cola Company, Telecommunication companies, have the need for complex software systems that can't be achieved by the current computer science and information technology students.

5.0 Goal and Objectives

5.1 Goal

The course aims to prepare students for careers in software engineering, software project management, and software development and integration

5.2 Objectives

The objectives of the programme are to:

- 1 Equip students with knowledge and skills in software engineering
- 2 Provide students with a comprehensive background in software engineering procedures and techniques.
- 3 Provide the theoretical and formal foundations to ensure precision in the software lifecycle.
- 4 Develop, within students, understanding of the importance of business acumen and communication skills in a typical Software Engineering environment.
- 5 Prepare students for lifelong learning

6.0 Learning outcomes

Upon successful completion of this program, the students will be able to:

- 1 Understand the analysis, synthesis and design of complex systems
- 2 Apply software engineering principles and techniques
- 3 Develop, maintain and evaluate large-scale software systems
- 4 Perform independent research and analysis
- 5 Apply the principles, tools and practices of software project management
- 6 Manage time, processes and resources effectively by prioritizing competing demands to achieve personal and team goals
- 7 Understand and meet ethical standards and legal responsibilities rapidly learn and apply emerging technologies
- 8 Demonstrate the phases in software development and explore the methods used in software development
- 9 Pursue further studies in a related field

7.0 Entry requirements

The general Admission requirements shall be decided and revised periodically by senate and made available by the academic registrar's office. The admission criteria of the BSc. SE shall consist of the following:

a) Admission by direct entry

Admission to the first year of the program will require a student to have sat the:

- Uganda Certificate of Education (UCE) and passed with at least a credit in mathematics in one sitting
- Uganda Advanced Level Certificate of Education (UACE) with at least two principal passes in science courses, with at least a grade D in mathematics; in one sitting

b) Diploma holders scheme

A person who holds a diploma in a related field or its equivalent from a higher institution of learning recognized by National Council of higher Education will be accepted. The prospective students are expected to start from year 1 with the direct entry students.

c) Mature Age Entry Scheme

To be eligible for the mature age scheme, the candidate must be at least 25 years old before the commencement of the program in the given academic year and must fulfill all the requirements of the mature age entry scheme of Mbarara University of Science and Technology.

8.0 Mode of delivery

The BSc. SE program will be conducted during the week from Monday to Friday starting from 8.00am to 5.00pm with a one hour lunch break. The program will employ a number of teaching and learning formats which include:

- Lectures, tutorial and practical sessions.
- In-class student presentations.
- Wide range of computer-based learning tools.
- The use of MUST e-learning system: <http://lms.must.ac.ug>
- case studies

8.1 Tuition fees

Privately sponsored students shall pay all the required fees to the University as determined by the senate and approved by council periodically. Ugandan students will pay tuition fees totaling to Two Million and Five Hundred and fifty two thousand shillings (2,110,500) per year. Foreign students will pay tuition fees of 3,120 US Dollars per year.

8.2 Program assessment

The general assessment of courses in this program will be based on of 100 total marks with proportions as:

- a) Continuous coursework consisting of 40 marks as broken down below;
 - I. Class Assignments 15 marks
 - II. Mid Semester Tests 15 marks
 - III. Group course work and presentation 10 marks
- b) Examination consisting of 60 marks. All this makes a total of 100%

However, some courses have varying assessment distributions that are described in the detailed course descriptions. A minimum of two course assignments/tests shall be required per course.

8.3 Grading system

Marks	80-100	75-79	70-74	65-69	60-64	55-59	50-54	0-49
Alpha grade	A	B+	B	C+	C	D+	D	F
GRADE POINT	5	4.5	4	3.5	3	2.5	2	0

A student who passes by at least 50% (grade point 2) gains all credit units allocated to the course. A student who scores less than 50% does not gain any credit. The student will be allowed to retake that course when it is next offered.

Classification of Degrees

Class	CGPA Range
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First Class	4.40– 5.00
Second Class (Upper Division)	3.60 – 4.39
Second Class (Lower Division)	2.80-3.59
Third Class (Pass)	2.00– 2.79

Weighting System

The weighting unit is a credit unit. One credit unit is one contact hour per week per semester. One contact hour can be defined as follows: -

- 1 lecture hour is equivalent to 1 contact hour.
- 2 tutorial hours are equivalent to 1 contact hour.
- 2 practical hours are equivalent to 1 contact hour.

All the students must make extensive use of the computing facilities outside the scheduled lecture, tutorial and practical hours. The details of the course structure is shown below, where LH, TH, PH, CH and CU stand for Lecture Hours, Tutorial Hours, Practical Hours, Contact Hours and Credit Units respectively.

8.4 Cumulative grade point average (CGPA)

The CGPA shall be calculated as follows: -

$$CGPA = \frac{\sum_{i=1}^n (GP_i * CU_i)}{\sum_{i=1}^n CU_i}$$

Where:

- GP_i is the Grade Point score of a particular course i ;
- CU_i is the number of Credit Units of course i ; and
- n is the number of courses so far done

8.5 Progression

8.5.1 Normal progress (NP)

This occurs when a student passes each course taken with a minimum Grade Point of 2.0.

8.5.2 Probationary progress (PP)

This is a warning stage and occurs if either the cumulative grade point average (CGPA) is less than 2.0 and/ or the student has failed a core course. Probation is waved when these conditions cease to hold.

8.5.3 Retaking the course (RT)

A Student may re-take any course when it is offered again in order to pass if the student had failed the course. A Student may take a substitute elective, where the Student does not wish to re-take a failed elective.

8.5.4 Discontinuation

When a student accumulates three consecutive probations based on the CGPA or the same core course(s), he/she shall be discontinued

9.0 Resources available to facilitate learning:

9.1 Human resource

The institute of computer science is blessed with academic staff who have diverse qualifications and skills to support students in different areas of computer science, information technology, computer and software engineering. Refer to Appendix

9.2 Physical infrastructures:

1. Lecturer rooms:

In the Kihumuro campus, the Faculty of computing and informatics has got spacious lecture rooms that can accommodate even up to 200 students in one sitting. More are under construction. The old campus has an additional lecture rooms to cater for all the needs of the faculty.

2. Computer labs (4) each with minimum of 40 computers
3. Library (2) each with minimum sitting capacity of 1000 students
4. E-learning facility – there is a Learning Management System

10. Graduation requirements

Software Engineering is a four –year program and a student must have covered at least 208 credit units to graduate.

SECTION B

CURRICULUM STRUCTURE (GENERAL OVERVIEW OF PROGRAM MATRIX)

Year one 1 semester 1

Code	Name of course	LH	PH	TH	CH	CU	status
SWE 1101	Discrete Mathematics	30	-	30	45	3	Core
SWE 1102	Structured Programming with C	45	30		60	4	Core
SWE 1103	Emerging trend in SE	45	-		45	3	Core
SWE 1104	Software development principles	45		30	60	4	Core
SWE 1105	Communication Skills	30	-	30	45	3	Core
SWE 1106	Fundamentals of software engineering	45		30	60	4	Core
DVS 1106	Introduction to political economy	45			45	3	Core
Total						24	

Year 1 semester 2

Code	Name of course	LH	PH	TH	CH	CU	status
SWE 1201	Object oriented programming 1	45	30		60	4	Core
SWE 1202	Data structures and algorithms	45	30		60	4	Core
SWE 1203	Operating Systems	45	30		60	4	Core

SWE 1204	Software Systems Engineering practice	45	-		45	3	Core
SWE 1205	Database programming	45	30		60	4	Core
SWE 1206	Computer architecture and organization	30		30	45	3	Core
DVS 1203	Political economy of Africa and Development	45			45	3	core
Total						25	

Year 2 Semester 1

Code	Name of course	LH	PH	TH	CH	CU	Status
SWE 2101	Object Oriented Programming II	45	30		60	4	Core
SWE 2102	Requirements engineering	45			45	3	Core
SWE 2103	OOP analysis and design(UML)	45	30		60	4	Core
SWE 2104	Algorithmic problem solving	45			45	3	Core
SWE 2105	Geographic information systems	45	30		60	4	Core
SWE 2106	Cloud computing	45			45	3	Core
SWE 2107	Internet technology and web design	45	30		60	4	Core
SWE 2108	Systems Analysis And Design	45	30		60	4	Core
DVS 2105	Political economy of Uganda	45			45	3	Core
Total						28	

Year 2 Semester 2

Code	Name of course	LH	PH	TH	CH	CU	status
SWE 2201	Formal specification for software engineering	45	-		45	3	Core
SWE 2202	Data communication Technology	45	30		60	4	Core
SWE 2203	Software Architecture and design	45			45	3	Core
SWE 2204	Software Metrics	45	-		45	3	Core

SWE 2205	User interface design and development	45	30		60	4	Core
SWE 2206	Distributed Processing	45	30		60	4	Core
SWE 2207	Industrial training I		120		60	4	recess
DVS 2201	Citizenry, Professionalism, Globalization and Entrepreneurship	45			45	3	Core
Total						28	

Year 3 Semester 1

Code	Name of course	LH	PH	TH	CH	CU	status
SWE3101	Human computer interaction	45	30		60	4	Core
SWE3102	Software reliability and testing	45	30		60	4	Core
SWE3103	Software project management	45		30	60	4	Core
SWE3104	Technical documentation for software engineers	45			45	3	Core
SWE3105	Modeling and simulation	45	15	15	60	4	core
SWE3106	Data Mining	45	30		60	4	Core
SWE3107	Mobile networks and Computing	45	30		60	4	Core
	Total					27	

Year 3 Semester 2

Code	Name of course	LH	PH	TH	CH	CU	status
SWE3201	Embedded systems software	45	30		60	4	Core
SWE3202	Mobile computing and Applications	45	30		60	4	Core
SWE3203	Software Engineering Industrial Min Project I	-	120		60	4	Core
SWE3204	Research methods in computing	45			45	3	core
SWE3205	Compiler construction	45	30		60	4	Core
SWE3206	Systems programming-Linux	45	30		60	4	Core
SWE3207	Industrial Training II	-	120			4	recess

	Total					27	
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Year 4 Semester 1

Code	Name of course	LH	PH	TH	CH	CU	Type
SWE4101	Computer Aided Design	45	30		60	4	Core
SWE4102	Computer Graphics Engineering	45	30		60	4	Core
SWE4103	Formal methods	45			45	3	Core
SWE4104	Software Design Patterns	45	30		60	4	Core
SWE4105	Artificial Intelligence	45	30		60	4	Core
SWE4106	Software Engineering Industrial Mini project II		120		60	4	Core
SWE4107	Client Server Programming for Applications	45	30		60	4	Core
Total						27	

Year 4 Semester 2

Code	Name of course	LH	PH	TH	CH	CU	status
SWE4201	Ethical and professional issues in Computing	45			45	3	Core
SWE4202	Software Maintenance and Evolution	45	30		60	4	Core
SWE4203	Entrepreneurship	45			45	3	Core
SWE4204	Software security	45	30		60	4	Core
SWE4205	Numerical Analysis	45	30		60	4	Core
SWE4206	Final Year Project	-	120			4	Core
Total						22	

YEAR ONE SEMESTER ONE

DISCRETE MATHEMATICS

Course Name
Course Level
Course Credit
Contact hours
Course code

DISCRETE MATHEMATICS
Year I Semester I
3CU
45 Hours
SWE 1101

Brief Course Description

This course introduces the foundations of discrete mathematics as they apply to computer science. It focuses on providing a solid theoretical foundation for further work. Topics covered include functions, relations, sets, simple proof techniques, Boolean algebra, propositional logic, digital logic, elementary number theory, and the fundamentals of counting.

Course Objectives

- 1 The course aims to provide students with:
- 2 Knowledge about operations associated with sets, functions, and relations with examples
- 3 Understanding of the basic counting principles, including uses of diagonalisation and the pigeonhole principle
- 4 Understanding of which proof is best for a given problem and the basic structure of each proof technique giving examples
- 5 The ability to compute permutations and combinations of a set, and interpret the meaning in context of particular applications
- 6 The knowledge require to Analyze a problem to create relevant recurrence equations or to identify important counting questions
- 7 An appreciation of the effect of AND, OR, NOT and EOR operations on binary data.
- 8 Identify basic methods of proof, particularly induction, and apply them to solve problems in mathematics and computer science;
- 9 Use and analyse simple first and second order recurrence relations;
- 10 Use trees and graphs to solve problems in computer science;

- 11 Apply counting principles in combinatorics;
- 12 Describe the principles of elementary probability theory, evaluate conditional probabilities and use Bayes' Theorem.

Learning Outcomes

Upon successful completion of the course, students shall be able to:

- 1 Explain the basic concepts of discrete structures and appreciate their importance as they apply to computing
- 2 Manipulate formulae involving sets, integers, reals and functions of such quantities
- 3 Solve simple problems involving sets, functions, graphs and trees
- 4 Construct sound logical arguments, including use of induction
- 5 Appreciate the way that discrete mathematics can assist problem solving

Course Outline

1	Introduction to Logic and Proofs	3 hours
2	Fundamental Structures	4 hours
3	Boolean Algebra	6 hours
4	Basic Logic	3 hours
5	Digital Logic	6 hours
6	Elementary Number Theory	4 hours
7	Basics of Counting	4 hours
8	Case study and Tutorials (30 hours)	15 hours

Mode of Delivery

Lectures, Tutorials and Group work

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

- 1 Rosen K. (2012) Discrete Mathematics and its Applications. (7th Ed) McGraw-Hill
- 2 Stein L. C., Drysdale, R. and Bogart, K. (2010) Discrete Mathematics for Computer Scientists. Addison-Wesley4

- 3 Balakrishnan, V.K. (1997). *Introductory Discrete Mathematics*. New York: Dover.
- 4 Bobrow, L.S. and Arbib, M.A. (1974). *Discrete Mathematics: Applied Algebra for Computer and Information Science*. Philadelphia, PA: Saunders.
- 5 Dossey, J.A. Otto, AD. Spence, L. and Eynden, C.V. (1997). *Discrete Mathematics*, 3rd ed. Reading, MA: Addison-Wesley.
- 6 Scheinerman, Edward, (2006). *Mathematics: A Discrete Introduction*, Cengage.
- 7 Roman, Steven. *An Introduction to Discrete Mathematics*, 2nd e

STRUCTURED PROGRAMMING WITH C

Course Name	Structured programming with C
Course Level	Year I Semester I
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 1102

Brief Course description

The course provides a strong base in the principles and practice of structured programming. A high level programming language such as C is used to explain the principles of programming and provide students with hands on practical skills. Areas covered include program structure, data structures, syntactical and semantic correctness, planning and segmentation in programming as well as working with data files.

Course Objectives

The course aims to provide students with:

- i) Knowledge about the various programming languages
- ii) Knowledge in basic programming concepts
- iii) Comprehensive knowledge about structured programming
- iv) Knowledge in planning and organization of programming projects
- v) Techniques of evaluating syntactic and semantic correctness of a computer program
- vi) Strong practical foundation in programming

Learning Outcomes

Upon Completion of the course, the students shall be able to:

- i) Explain the key differences between the various programming languages
- ii) Demonstrate understanding about the basic programming concepts
- iii) Build software using a functional programming language such as C
- iv) Plan and organize a programming project
- v) Evaluate a computer program for syntactic and semantic correctness

Course Outline

This course covers the following topics:

- | | | |
|---|---------------------------------------|---------|
| 1 | Introduction to programming languages | 4 hours |
| 2 | Program structure | 2 hours |
| 3 | Variables and Operators | 2 hours |

4	Conditional statements	5 hours
5	Looping statements	5 hours
6	Arrays and strings	5 hours
7	Functions	3 hours
8	Advanced data types	3 hours
9	Pointers	3 hours
10	Dynamic memory allocation and dynamic structures	3 hours
11	Working with files	3 hours
12	GUI	2 hours
13	Practical sessions (40 hours)	20 hours

Mode of Delivery

Lectures and Practicals

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

- 1 Bradley, L. J and Aitken, P. (2002) Sams Teach yourself C in 21 Days (6th Ed) Sams Publishing.
- 2 Deitel, P. and Deitel, H. (2012) C: How to program (7th Ed) Prentice Hall.
- 3 Perry, G. (2000) C by Example (4th Ed) Que Publishing.
- 4 www.cprogramming.com

EMMERGING TRENDS IN SOFTWARE ENGINEERING

Course Name	Emerging trends in Software Engineering
Course Level	Year 1semester 1
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 1103

Description:

The course is to expose provide students with an opportunity to search for knowledge in an area of interest. It is to allow a student do lightweight research and explore the current trends in a certain computer science areas

Course Aims and Objectives

- To build the research and problem-solving skills of the student by projecting the future trends in computer science.
- To identify computing solutions that can fit in the Ugandan context while remaining relevant to the international standards
- To aid a student get an in depth understanding of the developments in one area of computer science
- To empower the students with innovation and creative skills on how to implement their ideas

Course Learning Outcomes: After taking this course students should be able to:

- Discuss the various areas in computer science that are potential areas of research
- Demonstrate skills acquired in research methods
- Build a concrete idea ready for funding

Indicative Course content:

The content is not specific but will be dependent on the area the student chooses to pursue. The course will first introduce the students to the current trends in the context of Uganda and the global perspective. The lecturer will provide materials every week that students will read to gain more knowledge on the key topics bellow.

The major themes are but not limited to;

Current software applications in: Transportatio, Education,Health, Environment, Energy, Governance, Entertainment , Business [Business modeling and Financial analysis] The sub themes that could be looked at under each of the above themes include but are not limited to; Internet of things (IoT), Smart Grid systems, Software programming, Databases , Mobile programming , Computer design and architecture, Cloud computing Smart cities, Green Energy, Big data, Distributed systems, Computer networks, Computer hardware, Smart Cars

References No particular reference will be used for this course unit, except whatever is recommended by head of department in consultation with members of the department. Internet may be used.

Modes of assessment

Tests	20%
• Assignments	20%
• Examination	60%

SOFTWARE DEVELOPMENT PRINCIPLES

Course Name	Software Development Principles
Course Level	Year 1 semester 1
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 1104
Course Objectives	

The goal of this course is to study and apply advanced software engineering best practices, principles, and patterns that concern different phases of the software development lifecycle and different aspects of the software itself. In particular, we will study in detail the fundamental GoF object-oriented design patterns, object-oriented reengineering patterns, and patterns that facilitate the evolution of databases.

Course Objectives Upon successful completion, the student should be able to:

1. Demonstrate mastery of the software development process, describing core activities at various levels of software development;
2. Perform requirement analysis of a simple software systems;
3. Manage a team of developers; (iv) Demonstrate knowledge of at least one tool of software documentation;
4. Work in a team; and
5. Design, develop, test and valid a software product.

Course Content

Topics covered include:

Introduction: What is software engineering? Phases in the development of software. **12hrs**

Software Management: Planning and controlling a software development project. **12hrs**

Requirements analysis: Requirements specification, humans as information sources, tools for documenting the requirements (SADT, PSL/PSA, Ada-based analysis methods). **12hrs**

Software design: Abstraction, modularity, information hiding, design methods, design documentation. **12hrs**

Testing: Test objectives, verification and validation, manual test techniques, techniques for functional and structural testing. **12hrs**

Reading materials

1. Balancing Agility and Discipline: A Guide for the Perplexed; by Barry Boehm and Richard Turner, Addison Wesley Professional, 2003, ISBN-10: 0321186125.
2. 201 Principles of Software Development, by Alan M. Davis, McGraw-Hill, March, 1995, ISBN-10:

Modes of assessment:

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

COMMUNICATION SKILLS

Course Name COMMUNICATION SKILLS

Course Level Year I Semester I

Course Credit 3CU

Contact hours 45 Hours

Course code SWE 1105

Brief Course Description

This course provides students with skills to effectively communicate with their peers, subordinates and superiors. These skills include verbal, written, and non-verbal gestures. It improves their command of the English language in the areas of reasoning, writing and other expressions.

Course Objectives

The course aims to equip students with skills in:

- 1 Reading, writing and disseminating of information
- 2 Collecting and synthesizing information
- 3 Critical thinking and problem solving
- 4 Utilizing the library and other educational resources

Learning Outcomes

Upon completion of this course, the student shall be able to:

- 1 Effectively read, write and disseminate information
- 2 Collect and synthesize information
- 3 Generate solutions using critical thinking and problem solving skills
- 4 Utilize the library and other educational resources

Course Outline

The topics covered are in the broad areas of:

- | | |
|-------------------------------|-----------------|
| • Interpersonal skills | 6 hours |
| • Writing skills | 9 hours |
| • Reading skills | 6 hours |
| • Presentation skills | 6 hours |
| • Examination skills | 3 hours |
| • Tutorials and Presentations | 15 hours |

Mode of Delivery

Lectures, Tutorials, Presentations

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. Bovee, C.L., Thill, J.V., Schatzman, B.E. (2003). *Business Communication today*; Seventh Edition. New Dehli: Pearson Education
2. Cottrell, S. *The Study Skills Handbook* Fourth Edition. 2013
3. Fairbairn, G. J. & Winch, C. *Reading, writing and reasoning* (3rd ed. 2010) 428.40711 FAI
4. Alan Barker. *Improve Your Communication Skills Volume 39 of Creating Success Series*. Third Edition, Kogan Page, Limited, 2013

FUNDAMENTALS OF SOFTWARE ENGINEERING

Course Name Fundamentals of software engineering
Course Level Year 1 semester 1
Course Credit 4CU
Contact hours 60 Hours
Course code SWE 1106

Brief Course Description

This unit introduces the discipline of software engineering. The emphasis is upon a broad coverage of the areas, since students will at this early stage not have adequate programming skills to tackle many of the topics in greater depth. The notion of a software system as a model or approximation of a desired system is introduced, and used as a way of describing such things as the software life cycle and its various models, programming by contract, design and testing issues, maintenance, reuse, complexity, divide and conquer strategies, metrics and measurement, project management and software legacy.

Course objectives

- 1 To introduce software engineering and to explain its importance
- 2 To set out the answers to key questions about software engineering
- 3 To introduce ethical and professional issues and to explain why

Learning Outcomes

At the completion of this unit students should be able to:

- 4 Analyse a software engineering scenario and identify advantages and disadvantages of alternative SE life-cycle models;
- 5 Recognise and describe the components of a SE life-cycle model and apply them to a simple problem;
- 6 Produce and evaluate object-oriented software designs represented in a standard SE design formalism;
- 7 Perform basic black-box and white-box testing on a piece of software;
- 8 Describe and compare common SE team structures;
- 9 work collaboratively to carry out simple software engineering tasks;
- 10 Analyse a professional scenario with respect to the SE Code of Ethics and/or sustainability;
- 11 Synthesize an SE topic and present it to their peers.

COURSE Content

- **Introduction** This covers frequently asked questions about software engineering and professional/ethical responsibilities for software development **3hrs**
- **Software systems engineering** This will cover system environment, modelling, and procurement **3hrs**
- **Software processes** This will cover software models, design, specification, validation and evolution **4hrs**
- **Software project management** This will cover software project management practices, planning, scheduling, and risk management **6hrs**
- **Software requirements engineering** This will cover feasibility study, requirements elicitation, validation and management **6hrs**
- **Software prototyping** This will cover prototyping techniques, interfaces and processes
- **UML** this will cover use of Unified modelling Language [UML], to design use cases, activity diagrams... to demystify systems operational requirements **8hrs**
- **Software verification and validation** **3 hrs**
- **Managing legacy software systems** This will cover legacy systems structure, design and assessment **5hrs**
- **Software change and configuration management**• Software project management techniques: project metrics, software measurement and metrics, software quality metric. **5hrs**
- Risk management: software risks, risk identification, risk projection, risk mitigation, monitoring and management; Project Scheduling: people and effort relationships, defining tasks, defining task network, scheduling techniques; Software teams and intra-team relationships; role of project manager. **8hrs**
- Software Quality Assurance: Concept of quality, quality control vs. quality assurance, cost of quality, factors that affect quality, quantitative view of quality, quality metrics, defect removal efficiency SQA activities, ISO standards and CMM practices, SEI levels, Software reviews, Formal approaches to SQA, Statistical Quality Assurance. Software reliability, reliability metrics, reliability models, meeting reliability requirements. **9hrs**

Assessment

Examination (3 hours): 60%; In-semester assessment: 40%

Workload requirements

Minimum total expected workload equals 12 hours per week comprising

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Teaching Software Engineering through Simulation, Emily Oh and André van der Hoek
Institute for Software Research University of California, Irvine
2. Software engineering prospects over a decade from 2006 to 2016
3. Applying UML and Patterns. Craig Larman, 2nd *Edition*, *Prentice Hall*, 2002.
4. Software Engineering., Sommerville, 9th *Edition*, *Addison Wesley*, 2010
5. Software Engineering : A practitioner's Approach by Roger S. Pressman, 7th edition

INTRODUCTION TO POLITICAL ECONOMY

Course Name : Introduction to Political Economy

Course Level : Year 1 Semester 1

Course Code : DVS 1106

Credit Units : 3

Contact Hours: 45

Description

Political economy is concerned with the interaction of political and economic processes in a society i.e. the distribution of power and wealth between different groups and individuals, and the processes that create, sustain and transform these relationships over time.

The ways in which allocation and distribution of resources is done normally change over time, and therefore the course focuses on the laws that govern the development of the society, economy and power relations and various institutions that support them at each stage of human development and the subsequent social conflicts and institutions that are created to contain them. Political economy makes extensive and intensive use of class analysis in making sense of society and history, but does so in the context of political, cultural, and environmental processes, as well as other economic processes. It is concerned with understanding the role of various social economic systems in shaping and transforming society and history. The interdisciplinary nature of Political Economy is designed to offer students an exposure to the core concepts of political economy. The course focuses on investigating the relationship between the economy and politics as they affect the social and cultural life of societies, and secondly, focuses on the historical processes whereby social change is located in the interaction of the economic, political, cultural, and ideological moments of social life.

Objectives:

The primary objective of the course is for students to critically evaluate the literature on political economy, students will gain an appreciation for the flaws and limits of this literature, for the nature of economic debate and the manner in which particular strands of this literature evolve, for inherent empirical challenges that arise in attempting to assign causality among correlated economic and political variables, and for the statistical techniques commonly employed to address these challenges

Learning Outcomes

At the end of the course students should be able to:

- Understand relations between economy and political power and the ways in which economic policy are shaped by political institutions
- Understand that there are laws governing the development of societies
- Comprehend the transformation processes that are embedded in the society transformation from socio-economic system to another and understand the basics of conducting class analysis

- Appreciate the dynamics of development issues today and as future policy makers, start to formulate their own views
- Apply the knowledge acquired on societal transformation to contemporary socio-economic debates

Course Outline

- Key terms in Political Economy, **4hrs**
- Historical development of political economy **4hrs**
- Why study Political Economy? **4hrs**
- Methodological approach to the study of society **4hrs**
- Hegel Philosophy & Karl Marx's Historical and Dialectical Materialism **4hrs**

Socio-economic systems 20hrs

- Primitive Communalism
- Slavery
- Feudalism
- Capitalism- Mercantilism, Industrial Capitalism, Imperialism
- Socialism-Socialism, Social Democracy and Democratic Socialism
- Communism

Introduction to Theories of International Political Economy- 5hrs

- Realism
- Liberalism
- Marxism

Mode of delivery

Interactive lectures, tutorials, case studies, group work – students based collaborative learning.

Mode of assessment

1. Continuous assessment (30%)
 - a. Tests (20%)
 - b. Presentations – case studies (5%)
 - c. Group work and participation (5%)
2. Final written examination (70%). The final examination will be similar across Faculties, administered on the same day.

Suggested Readings

Heywood Andrew. (2000). *Key Concepts in Politics*. New York: Palgrave

Heywood Andrew. (2002). *Politics*. New York: Palgrave

Sen Gautam. (2005). "The political economy of Development" in Haynes, J.,ed (2005) *Development Studies*. England: Palgrave Macmillan pp 89 – 115

WadadaNabudere. (1978). *The Political Economy of Imperialism*. Dar es Salaam: Tanzania Publishing House

Bowles Samauel, Richard Edwards, Frank Roosevelt. (2005). *Understanding Capitalism, Competition, Command and Change*. New York and Oxford: Oxford University press.

Cutler Anthony, Hindess Barry, Hirst Paul & Hussain Athar. (1977). *Marx's Capital and Capitalism Today*, Vol I. London: Routledge & Kegan Paul

Clilote R. (1990). *The Political Economy of Imperialism: Critical Appraisals*. Kuwer: Academic Publishers.

Nabudere, D. W. (1980). *Imperialism and Revolution in Uganda*. Dar es Salaam: Tanzania Publishing House

Mandel Ernest, 4th Edition. (1977). *Marxist Economic Theory*. London: The Merlin Press

YEAR ONE SEMESTER TWO

OBJECT ORIENTED PROGRAMMING 1

Course Name **OBJECT ORIENTED PROGRAMMING**

Course Level **Year 1 SEMESTER 2**

Course Credit **4CU**

Contact hours **60 Hours**

Course code **SWE 1201**

Course Description

The object oriented programming paradigm has been used in many projects to build complex software systems. This course provides students with the opportunity to acquire skills for building software based on the object oriented programming paradigm. C++ and/or Java are some of the languages used in this course.

Course Objectives

The course aims to enable students to:

1. Understand and use the Object Orientation Paradigm for software development
2. Develop software using a programming language like C++ and/or Java
3. Understand the use of basic Java features in a working program
4. Understand the use of the following advanced features of inheritance, encapsulation, overloading, polymorphism, abstract classes and interfaces in a program
5. Develop knowledge in the use of GUI-based event-driven programming in a working program assignment utilizing GUI components, event-listeners and event-handlers

Learning Outcomes

Upon successful completion of the course, students shall be able to:

1. Demonstrate the understanding and application of Object oriented design techniques for software development using a programming language like C++ and Java
2. Demonstrate correct use of the basic features in a working program: objects, classes, methods, IO handling, decisions and iterations
3. Demonstrate correct use of the following advanced features in a working program: inheritance, encapsulation, overloading, polymorphism, abstract classes and interfaces

4. Demonstrate knowledge of GUI-based event-driven programming in a working program assignment utilizing GUI components, event-listeners and event-handlers

Course Outline

Topics include:

- | | |
|--|---------|
| 1. Theory of the object oriented paradigm | 4hours |
| 2. Data types, inheritance, exceptions, templates and patterns | 3hours |
| 3. Object Oriented programming languages (e.g. C++ and Java) | 8hours |
| 4. Encapsulation, inheritance, and polymorphism | 4hours |
| 5. Abstract data types, concrete data types and subtypes | 3hours |
| 6. Virtual functions and operator overloading and pointers | 4hours |
| 7. Object Oriented computing in a distributed systems, applets | 3hours |
| 8. Classes, interfaces and packages | 4hours |
| 9. Selection and application of current design and development tools | 4hours |
| 10. Practical Lessons | 22hours |

Mode of Delivery:

Lectures, Tutorials, Practicals, Group work

Course Assessment

- | | |
|---|--------|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |
| • Virtual functions and operator overloading and pointers | 4hours |

Reading materials

1. John Hubbard, Programming in C++ McGraw Hill Schaum's Outline Series, 2000.
2. Y. Daniel Liang, Introduction to Programming with C++ (Brief Version), Prentice Hall, 2007.

DATA STRUCTURES AND ALGORITHMS

Course Name	DATA STRUCTURES AND ALGORITHMS
Course Level	Year I Semester 2
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 1202

Course Description

This course exposes students to data structures and corresponding algorithms. It also equips them with the skills to measure the performance of an algorithm as well as the techniques on how they can be further improved.

Course Objectives

This course aims at equipping students with knowledge about:

- 1 The different types of data structures and identify their appropriateness
- 2 Algorithms and their different representations in terms of mathematical functions or formulae
- 3 The performance and management of simple algorithms
- 4 Cost-effective algorithms

Learning Outcomes

On completion of course, the student shall be able to:

- 1 Use the different types of data structures
- 2 Perform an algorithm analysis and represent algorithms in terms of mathematical functions
- 3 Evaluate the performance and management of simple algorithms
- 4 Develop cost-effective algorithms

Course Outline

The topics covered in this course include:

- | | |
|---|-----------------|
| 1 Elementary data structures | 6 hours |
| 2 Advanced data structures | 10 hours |
| 3 Trees, graphs, networks, functions, signals | 8 hours |
| 4 Sorting and searching | 5 hours |
| 5 Analysis of algorithms | 5 hours |
| 6 Recursion | 3 hours |

7 Algorithm performance and optimization

8 hours

8 Practicals (30 hours)

15 hours

Mode of Delivery

Lectures, tutorials and practicals

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

- Data Structures and Algorithm Analysis in C, by Mark weiss, 2nd Edition, Addison-Wesley, (1997), ISBN: 0-201-49840-5
- Data Structures and Algorithm Analysis in Java, by Mark Allen Weiss, Pearson International, ISBN 0-321-37319-7

OPERATING SYSTEMS

Course Name	OPERATING SYSTEMS
Course Level	Year I Semester 2
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 1203

Course Description

This course introduces learners to the services of operating systems. It exposes them to details on how an operating system controls all the processes of a computer system including those of peripheral devices.

Course Objectives

The aims of this course are to:

- 1 Create understanding about the role of operating systems in the management of computer system resources
- 2 Provide in-depth knowledge of operating system processes and principles
- 3 Enable students to understand computer system performance

Learning Outcomes

Upon completion of this course, the student shall be able to:

1. Have a proper understanding of the differences between these two operating systems
2. Understand different design principles for operating systems and various software tools that make operating systems usable.
3. be familiar with the use of data structures as the foundational base for computer solutions

Course Outline

1	Operating System concepts	3 hours
2	Structure of an operating system	6 hours
3	System processes	9 hours
4	Threads	6 hours
5	Deadlocks	6 hours
6	Memory management	9 hours
7	Device management	6 hours

Mode of Delivery

Lectures and Tutorials

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. Abraham Silberschatz and Peter Baer Galvin. (1998) *Operating System Concepts*, Addison-Wesley, (5th ed.)
2. Andrew S. Tanenbaum and Albert S. Woodhull. (1997) *Operating Systems: Design and Implementation*, Prentice Hall, (2nd ed.)
3. Andrew S. Tanenbaum. (2007) *Modern Operating Systems* Prentice Hall, (3rd Edition)
4. d. Thomas W. Doeppner, (2010) *Operating Systems In Depth: Design and Programming*, Wiley, (1st Edition)

SOFTWARE SYSTEMS ENGINEERING PRACTICE

Course Name SOFTWARE SYSTEMS ENGINEERING PRACTICE

Course Level Year 1 Semester 2

Course Credit 3CU

Contact hours 45 Hours

Course code SWE 1204

Brief Course Description

This course examines software as an engineering discipline. It explains the importance of systems engineering and elaborates the related concepts of systems and modern engineering. The emphasis is on complex software systems, requiring a high degree of rigor and formality.

Course Objectives

The aims of this course are:

- 1 To introduce students to the systems engineering approach to software systems
- 2 To equip students with analytical skills so they can evaluate the factors leading up to system complication and complexity, the causes of a systems failure, the importance of stakeholders involvement in a systems engineering project and using input-output diagramming
- 3 To enable students appreciate the importance of systems engineering important as well as the difference between process science and engineering
- 4 To impart knowledge of a systems engineering project in terms of the balance of demands, choice and constraints
- 5 To equip students with an understanding of the lag involved in controlling a system as well as the differences between a generic model of systems engineering and a software specific model

Learning Outcomes

Upon successful completion of this course students will be able to:

- 1 Classify new software product developments as: fault correction, enhancements, new but similar products, radically different, revolutionary or iconoclastic products
- 2 Analyse the causes of a systems failure
- 3 Identify and evaluate the relationships of the factors leading up to system complication and complexity

- 4 Answer the question ‘why is systems engineering important?’
- 5 Define the difference in process between science and engineering
- 6 Identify and analyse the importance of stakeholders involved in a systems engineering project
- 7 Classify a systems engineering project in terms of the balance of demands, choice and constraints;
- 8 Analyse a system using input-output diagramming
- 9 Estimate the lag involved in controlling a system
- 10 Evaluate the differences between a generic model of systems engineering and a software specific model

Course Content

1 Characteristics of systems engineering	3 hours
2 Challenges in a systems engineering project	3 hours
3 Scope of a systems engineering problem	3 hours
4 Identify the stakeholders and other factors that shape the system requirements	4 hours
5 Emergent system properties	3 hours
6 Identifying the real problem which the system is intended to solve	4 hours
7 Technological, operational and economic considerations in the design process	4 hours
8 Interactions between a system and its environment	3 hours
9 Integrating the system with existing systems	3 hours
10 Tutorials (30 hrs)	5 hours

Mode of Delivery

Lectures, Tutorials, Seminars, Group work

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. This Systems Engineering Principles and Practice (Wiley Series in Systems Engineering and Management) by Alexander Kossiakoff Hardcover £93.48
2. Modern Methods of Systems Engineering: With an Introduction to Pattern and Model Based Methods by Joe Jenney Paperback £12.79
3. Software Systems Engineering: Andrew P. Sage, James D. Palmer . ISBN-13: 978-0471617587 , Wiley-Interscience; 1 edition (March 1990)
4. Systems Engineering Principles and Practice, Second Edition Alexander Kossiakoff, William N. Sweet, Samuel J. Seymour, Steven M. Biemer, rint ISBN: 9780470405482, Online ISBN: 9781118001028

DATABASE PROGRAMMING

Course Name DATABASE PROGRAMMING

Course Level Year I Semester II

Course Credit 4CU

Contact hours 60 Hours

Course code SWE 1205

Brief Course Description

This course introduces the student to the basic theoretical and practical concepts of a database, its setup, implementation, use and maintenance in a typical business organization.

Course Objectives

The course aims to provide students with:

- 1 The theory of database systems and data modeling
- 2 Practical skills in developing secure relational databases
- 3 Knowledge to formulate queries in databases
- 4 Knowledge of at least one DBMS
- 5 Skills to import and export data to and from different applications

Learning Outcomes

Upon completion of the course, the student shall be able to:

- 1 Demonstrate understanding of the basic concepts of database systems and data modeling
- 2 Develop secure relational databases
- 3 Formulate queries in databases and to use at least one DBMS
- 4 Import and export data to and from different applications

Course Outline

The topics to be covered include:

- | | | |
|---|----------------------------------|----------------|
| 1 | Introduction to database systems | 3 hours |
| 2 | Database construction principles | 6 hours |
| 3 | Database architecture | 3 hours |
| 4 | Data modeling | 9 hours |
| 5 | Database querying | 6 hours |
| 6 | Database Security | 6 hours |
| 7 | Database Managements Systems | 6 hours |

8 Distributed databases

6 hours

9 Practical lessons (30 hours)

15 hours

Mode of Delivery

Lectures, case studies and Practicals

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Ramez. E, Shamkant B. N, Database System: Models, Languages, design and Application Programming. 6th Edition
2. Peter .R, Carlos C, Keeley .C, Database Systems: Design, Implementation and Management
3. Abraham. S, Henry F. K, Sudarshan S, Database System Concepts, 5th Edition
4. Micheal .M, Oracle Database: 11g – PL/SQL Programming
5. Micheal. V, Database design, Application, Development and Administration, 3rd Edition

COMPUTER ARCHITECTURE AND ORGANISATION

Course Name	COMPUTER ARCHITECTURE AND ORGANISATION
Course Level	Year 1 Semester 2
Course Credit	3CU
Contact hours	45 Hours
Course code	SWE 1206

Brief Course Description

This course introduces students to the organization and architecture of a computer. It explores the composition of a computer including its design, controlling functionality, processing and memory management.

Course Objectives

The course aims to:

- 1 Introduce the concepts in computer organization
- 2 Explain the architectural features of a modern computer
- 3 Provide understanding of the principles governing the interaction between hardware and software components
- 4 Describe the relationship between the different components and their roles in the operations of the computer

Learning Outcomes

On completion of this course, the student shall be able to explain the:

- 1 Relevant concepts in computer organization
- 2 Architectural features of a modern computer system including its design and development
- 3 Principles governing the interaction between hardware and software components
- 4 Relationship between the different components and their roles in the operations of the computer

Course Outline

The major topics covered include:

- | | |
|--|----------------|
| 1 Digital logic | 6 hours |
| 2 Memory management | 6 hours |
| 3 Computer processor architectures | 6 hours |
| 4 Machine Language and CPU Instructions sets | 6 hours |

5	Instruction processing	3 hours
6	I/O Programming	3 hours
7	Tutorials	15 hours

Mode of Delivery

Lectures and Tutorials

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Computer Systems Architecture by M. Morris Mano, Prentice Hall, 1993.
2. Structured Computer Organization by Andrew S. Tanenbaum, Prentice Hall 1984.
3. Computer Systems Concepts and Design by Glenn B. Gibson, Prentice Hall, 1991
4. Computer Organization and Architecture by William Stallings, Prentice Hall 2003.

POLITICAL ECONOMY OF AFRICA AND DEVELOPMENT

Course Name: Political Economy of Africa and Development

Course Level: Year 1 Semester 2

Course Code: DVS 1203

Credit Units: 3

Description

The aim of this course is to increase students' capacity to use their growing knowledge of political economy of development in rigorous analysis of a subset of critical issues facing policy makers in sub-Saharan Africa. The African continent stands high among the many countries facing development challenges. The exposure to which Africa received from Europe during colonialism; the post-colonial situation and the current globalization drive leave Africa in a development dilemma. African development has gained prominence in the development discourse. Every development studies service course student that is interested in understanding the mechanisms and processes through which social transformation can be created needs to understand the position and processes through which Africa has transformed. The course interrogates African development problem taking into consideration the forces that shaped the colonization of Africa; colonialism and its social, economic and political impact on Africa

Learning Objectives By the end of the course, students should:

1. Have a critical and broad understanding of African politics and the historical forces that have shaped its reality.
2. Be conversant with the dominant debates in African politics and have the conceptual and theoretical language to critically analyse politics on the continent;
3. Understand the relationships between global power and its implications for Africa;
4. Be able to place contemporary African politics in larger historical frameworks and understand the forces that continue to shape the continent and its present day reality.
5. Have a critical understanding of the power-knowledge regimes that shape discourses on and about African politics
6. Appreciate the historical connections between Africa and the world as well as its place in the world.

Learning Outcomes

At the end of the course students should be able to

- Understand development challenges that face Africa today from the historical and contemporary perspectives
- Suggest viable recommendations for Africa's development challenges from a well-informed perspective
- Apply their knowledge in constructing a critical analysis of the issues for a particular African country

Course Outline

Pre-colonial, Colonial and Postcolonial Africa

20hrs

- Pre-colonialism- Kingdoms, socio-economic arrangements, class, pre-colonial economy

- Slave Trade
- Colonialism- Actors, colonial economy, production- colonial economies-settler economy, plantation economy and peasantry economies, Regional Division of labor
- Resistance to Colonialism in Africa: resistors and Collaborators, Passive and Active Resistances (Maumau, Nama-Herero, MajiMajietc)
- National Movements (and personalities) and Fight for Independence

Regionalism and African Integration **5hrs**
 OAU/AU, SADC, ECOWAS, COMESA, EAC

Development and Underdevelopment **20hrs**

- Theories of Development
- Approaches to development: Top-Bottom, Bottom-up approaches
- Causes/factors and actors of underdevelopment
- Poverty: Poverty analysis, causes, dimensions, way forwards
- Democracy, good governance, human rights and Development

Mode of delivery

Interactive lectures, tutorials, case studies, group work – students based collaborative learning.

Mode of assessment

1. Continuous assessment (30%)
 - a. Tests (20%)
 - b. Presentations – case studies (5%)
 - c. Group work and participation (5%)
2. Final written examination (70%). The final examination will be similar across Faculties, administered on the same day.

Suggested Readings

Rodney Walter. (1972). *How Europe underdeveloped Africa*. London: Zed Books

Daron Acemoglu and James Robinson (2012). *Why Nations Fail: The Origins of Power, Prosperity and Poverty*

Boahen, A. A. (1987). *African Perspective on Colonialism*. Baltimore, Maryland

Mamdani, Mahmood. (1996). *Citizens and Subject: Contemporary Africa and the Legacy of Late Colonialism*. Princeton, NJ: Princeton University Press

Omotola, J. Shola & Saliu, Hassan. (2009). “Foreign aid, debt relief and Africa’s development: problems and prospects,” *South African Journal of International Affairs*, Vol. 16, Issue 1, pp. 87-102

Moyo, Dambisa. (2009). *Dead Aid: Why aid is not working and how there is a better way for Africa*, New York: Farrar, Straus & Giroux

Nkrumah, K. 1965, *Neo-colonialism: the last stage of Imperialism*, London: Heinemann

Nyerere, J.K. 1997, ‘Africa Must Unite’, edited excerpts from a public lecture delivered in Accra to mark Ghana’s fortieth Independence Day anniversary celebrations, United New Africa Global Network website: <http://www.unitednewafrica.com/Africa%20Unite.htm>

Museveni, Y. K (1992) What is Africa’s Problem? Kampala: NRM Publishers

Kothari Uma and Minogue Martin (Eds), (2002); *Development theory and practice*, Palgrave, London.

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Ocheni Stephen & Nwankwo Basil C. (2012). Analysis of Colonialism and Its Impact in Africa. Canadian Academy of Oriental and Occidental Culture, Vol. 8, No. 3, 2012, pp. 46-54

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Kanogo, Tabitha. *Squatters and the Roots of Mau Mau*. Athens: Ohio University Press, 1987.

Amartya Sen (1999) *Development as freedom*, Oxford: OUP

Willis Katie. (2005). *Theories and Practices of Development*. London: Routledge –Taylor and Francis

YEAR TWO SEMESTER ONE

OBJECT ORIENTED PROGRAMMING-JAVA II

COURSE NAME: OBJECT ORIENTED PROGRAMMING-JAVA II

Course Level: Year Two, Semester One

Credit Units: 4

Course Code: SWE 2101

Description

This course runs for one semester, i.e. fifteen weeks of teaching. This course makes students to have lecture-and-lab sessions to understand database-intensive, desktop and web-application development, Networking, file handling among others. Students doing this course are expected to have basic knowledge in introduction to object oriented programming with java I, and Introduction to Programming using C or any other equivalent language. Key topics include packages, file handling, JDBC database access, Networking, JavaServer Faces (JSF), Ajax enabled JSF components, building web services and consuming web services. Students will participate in extensive hands-on laboratory assignments.

Aims/Objectives

Prepare students to be able to:

- Employ more advanced concepts in Object Oriented Programming including Java database connectivity, multi-threading, handling HTTP requests and using the Model View Controller.
- Utilize case study problems to solve using advanced OOP techniques with specific emphasis on the use of threads, networking and remote method invocation.
- Develop Web applications by using Java Server Pages and Servlets

Course Learning Outcomes

Upon completion of this course, students will be able to:

- Document and package a Java application.
- Utilize many of the new enhancements added to the Java API.

- Develop robust Java code.
- Capture configuration and debugging information using the Java Logging APIs.
- Serialize Java objects.
- Develop TCP/IP Client Server applications using sockets. • Execute methods on a remote object using RMI.
- Perform database queries and updates using JDBC

Course Content

- Packaging and Distributing a Java Application *(8 hours)*
- File IO *(6 hours)*
- Multithreading *(8 hours)*
- Accessing Databases with JDBC *(8 hours)*
- Web Basic HTML/CSS/JavaScript *(8 hours)*
- Java Web Applications: Part 1 *(8 hours)*
- Web Applications: Part 2 *(6 hours)*
- JAX-WS Web Services *(8 hours)*

Mode of Delivery:

- Lectures
- Group Discussions
- Tutorials
- Lab Sessions
- Case Studies
- Projects

Methods of Assessment

Assessment will be in form of

- Continuous Assessment (40%)
 - Individual and Group Assignments
 - Mid Semester Tests
- Final Examination (60%)

Key Reading List

- Kathy, S. and Bert B. (2005) Head First Java (2nd Ed) O'Reilly
- Deitel, P. and Deitel, H. (2011) Java How to Program (9th Ed) Prentice Hall
- Eck, D. J. (2009) Programming: Introduction to Programming Using Java.

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- Moffat, A., (2003) Programming, Problem Solving, and Abstraction. Pearson
- Kenneth, C. L.(1993) Programming Languages: Principles and Practice, (2nd Ed) Thomson Brooks/Cole.

REQUIREMENTS ENGINEERING

Course Name
Course Level
Course Credit
Contact hours
Course code

REQUIREMENTS ENGINEERING
Year 2 SEMESTER 1
4CU
45 Hours
SWE 2102

Course Description

Requirements engineering is a software engineering process which covers all of the activities involved in discovering, documenting and maintaining a set of requirements for a computer-based system. It provides students with principles, tools techniques for requirements elicitation, specification, modelling, validation and management.

Course Objectives

The course aims to provide students with:

1. Knowledge about principles, tools and techniques for requirements elicitation, specification analysis and modelling
2. Understanding of the role of requirements in system development and maintenance
3. Skills in specifying requirements for Real Systems
4. Knowledge about managing changes to requirements as the system is being developed and put into use.

Learning Outcomes

Upon successful completion of the course students should be able to:

1. Apply the principles, tools, and techniques for requirements elicitation, specification, and analysis and modelling to a software engineering problem
2. Demonstrate the role of requirements in system development and maintenance
3. Explain the difficulties of specifying requirements for real systems, as well as effective methods tools and techniques.
4. Manage changes to requirements as the system is being developed and put into use

Course Outline

- | | |
|------------------------------------|--------|
| 1. Software Requirements Concepts, | 3hours |
| 2. Requirements Elicitation | 6hours |

3. Software Requirements Analysis,	6hours
4. Software Requirements Specifications,	6hours
5. Software Requirements Tools,	6hours
6. Software Requirements Verification,	6hours
7. Software Requirements Engineering Management,	6hours
8. Developing a Successful Software Requirement	6hours

Mode of Delivery

Lectures, Case studies, Group work

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

1. Mastering requirements process 2nd edition, by suzanna Robertson, james Robertson.
2. Requirements engineering by ian Sommerville and Pet S, johwilley and sons publishers
3. Software and systems requirements engineering in practice

OBJECT-ORIENTED ANALYSIS AND DESIGN (UML)

Course Name **OBJECT ORIENTED ANALYSIS AND DESIGN (UML)**
Course Level **Year 2 Semester I**
Course Credit **4CU**
Contact hours **45 Hours**
Course code **SWE 2103**

Brief Course Description

The course covers the important themes of object orientation and UML modeling, critical for the design and development of systems using the object oriented programming languages. Particular emphasis is laid on structure and behavioral modeling as well as object interactions.

Course Objectives

The course aims to provide skills on:

- 1 The concepts and advantages of object oriented analysis, design and its application to systems development.
- 2 Basic object-oriented concepts of; abstraction, encapsulation, inheritance, hierarchy, modularity, and polymorphism
- 3 Creating models used in object oriented programming languages such as C++ and Java
- 4 Iterative, use case-driven process to the development of a robust design model
- 5 Use of UML as a tool or language to represent the design model

Learning Outcomes

Upon Completion of the course, students shall be able to:

- 1 Explain the concepts and advantages of object oriented analysis and design for application systems development.
- 2 Apply Object Oriented concepts such as abstraction, encapsulation, inheritance, hierarchy, modularity, and polymorphism to the development of robust systems models
- 3 Represent design models using the Unified Modeling Language

Course Outline

The course covers the following main themes and associated topics:

- 1 Objects and classes 2 hours
- 2 Principles of the Object orientated paradigm (Abstraction, encapsulation, inheritance and hierarchy, modularity and polymorphism) 6 hours
- 3 UML modeling and design

a. Use case diagrams	6 hours
b. Structural diagrams.	6 hours
c. Interactions diagrams	6 hours
d. Deployment diagrams	3 hours
e. Component Diagrams	3 hours
4 UML relations: associations, aggregation, composition	3 hours
5 Tutorial and Case Study	20 hours

Mode of Delivery

Lectures, Case studies, Tutorials and Group work

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

Object oriented analysis and design by John Deacon, Addison Wisely publishers 2005, ISBN 0321263170

ALGORITHMIC PROBLEM SOLVING

Course Name	Algorithmic problem solving
Course Level	Year 2 semester 1
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 2104

Brief Course Description

Algorithms are recipes for solving a problem. They are fundamental to computer science and software engineering. Algorithms are the formal foundation of computer programming but also exist independently of computers as systematic problem-solving procedures. This unit introduces algorithms, the study of algorithms. It is not about programming and coding but rather about understanding and analysing algorithms and about algorithmic problem-solving, i.e. the design of systematic problem-solving procedures. The unit will not require any knowledge of a programming language and is very hands-on. Students will develop algorithms to solve a wide variety of different problems, working individually as well as together in groups and as a class.

Objective: this course aims to:

1. Introduce students to structured programming and design
2. Enable students explain the role and purpose of algorithms in software Engineering
3. Indicate how algorithms are part of many different computer applications
4. Provide some knowledge themes such as relating complexity with algorithms

Learning Outcomes

At the successful completion of this unit, students should be able to:

1. Describe an algorithm consisting of basic structures (sequence, choice, iteration, modules) at the level of detail required for a particular audience;
2. Demonstrate how basic data structures (list, graphs, trees, sets, tables) function;
3. Create simple recursive and iterative algorithms;
4. Evaluate different possible strategies for developing an algorithm and be able to select an appropriate one to solve a given problem;
5. Apply standard patterns to develop algorithms;

6. Break problems down into simpler problems;
7. Determine the complexity of simple algorithms;
8. Recognise the limitations of algorithms.

Algorithmic Problem Solving studies the implementation and application of algorithms to a variety of problems. Topics will include:

- programming basics: I/O, variable types and ranges **3hrs**
- time/space complexity analysis: estimate the execution time of a program **3hrs**
- fundamental data structures: array, linked list, stack, queue, binary search tree, hash table, heap **5hrs**
- tree data structures: union-find, trie **3hrs**
- problem solving strategies: complete search, greedy, binary/ternary search, divide-and-conquer **3hrs**
- decomposition and range queries: square-root/binary decomposition, segment tree, Fenwick tree, sparse table, heavy-light tree decomposition, lazy propagation **5hrs**
- graph algorithms: shortest path, topological sort, minimum spanning tree, Eulerian path/cycle, strongly connected components, articulation points & bridges **3hrs**
- flow and matching: maximum flow, bipartite matching **3hrs**
- dynamic programming (DP): subset sum, weight balancing, knapsack, longest common subsequence (LCS), edit distance, longest increasing subsequence (LIS), Hamiltonian path/cycle, advanced DP state design **8hrs**
- string matching: Rabin-Karp, polynomial string hashing, KMP **3hrs**
- mathematics: prime, sieve of Eratosthenes, combinatorics, probability and expected value **3hrs**
- computational geometry: turning direction, line/segment intersection, point in polygon, polygon area, convex hull **3hrs**

Course Assessment

- | | |
|-------------------------|-----|
| • Practical Tests | 20% |
| • Practical Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Data Structures and Algorithm Analysis in C, by Mark weiss, 2nd Edition, Addison-Wesley, (1997), ISBN: 0-201-49840-5
2. Data Structures and Algorithm Analysis in Java, by Mark Allen Weiss, Pearson International, ISBN 0-321-37319-7
3. Steven and Felix Halim's Competitive Programming (Third Edition)

GEOGRAPHICAL INFORMATION SYSTEMS

COURSE NAME: GEOGRAPHICAL INFORMATION SYSTEMS

Course level: year two, semester one

Credit units: 4

Course code: SWE 2105

Course Description

GIS (Geographic Information Systems) is a computer-based tool that uses spatial (geographic) data to analyze and solve real-world problems. This course is designed to introduce the student to the basic principles and techniques of GIS. The course is designed to provide the students with an understanding of the methods and theories of spatial analysis that will allow students to apply GIS knowledge and skills to everyday life. The lab material will emphasize GIS data collection, entry, storage, analysis, and output using ArcGIS.

Course Objectives

To prepare students to:

- Identify concepts, structures and functionalities of GISs.
- Employ common GIS techniques to collect, analyze, process, and present spatial or geographic data.
- Demonstrate the use of Global Positioning System, remotely sensed data, and other online GIS data sources for spatial analysis and mapping
- Demonstrate use of major commercial GIS packages such as GeoMedia and ESRI's ArcGIS software in GIS related industries.

Learning Outcomes

By the end of this course, the student will be able to:

- Identify concepts, structures, and functionalities of GISs

- Employ common GIS techniques to collect, analyze, process, and present spatial or geographic data.
- Utilise Global Positioning System, remotely sensed data, and other online GIS data sources for spatial analysis and mapping
- Design and present a GIS project that can do data capture, data storage, analysis, and presentation.
- Employ commercially available GIS software and seeking entry level position in GIS related industries.

Course Outline

- Introduction to GIS: the ACC GIS Program, the GIS profession, and GIS Software (**4 hours**)
- Introduction to the concept of envisioning information: how they apply to maps, symbolize features and rasters in ArcGIS, classify features and rasters in ArcGIS. (**4 hours**)
- Features and Attributes: model geospatial features, feature description, how features and attributes are linked and displayed in a GIS (**6 hours**)
- Attribute queries: selection methodologies available in ArcGIS, SQL to execute standard database queries, Creation of summary reports based on attribute queries (**4 hours**)
- • Tables: basic structure, type, and format for tables stored and accessed in a GIS (**4 hours**)
- Spatial queries: spatial relationships and how to query them in GIS, queries definition, multi-step spatial query, join attributes by location (**4 hours**)
- Geoprocessing: preparation and analyze data (**6 hours**)
- Spatial analysis: vector, raster, and three dimensional spatial analysis, spatial analyses using vector and tabular data (**4 hours**)
- Map projection and scale: coordinate systems, the types of errors (**3 hours**)
- Data Types, Structures and Formats (**3 hours**)

- Data Creation, Collection and Quality (*6 hours*)
- Geocoding: address locator, Interpret geocoding results (*4 hours*)
- Cartography: Basic cartographic techniques (*4 hours*)
- presentation medium and Models, metadata, and mapping law (*4 hours*)

Mode of Delivery

Lectures, Practical, peer discussions, presentations

Course Assessment

Coursework / continuous assessment (40%)

Final examination (60%)

Reading List

Recommended Textbooks: O'Sullivan, D. and D. Unwin. 2010. *Geographic Information Analysis*. 2nd Edition. John Wiley and Sons.

GIS Fundamentals: A First Text on Geographic Information Systems (2nd edition), by Paul Bolstad, published by Eider Press.

Getting to Know ArcGIS Desktop (2nd edition), by Tim Ormsby et al., published by Environmental Systems Research Inc. (ESRI).

CLOUD COMPUTING

COURSE NAME: CLOUD COMPUTING

Course Level: Year Two, Semester ONE

Credit Units : 3

Contact Hours: 45

COURSE CODE: SWE 2106

Brief description

This course gives students an overview of the field of Cloud Computing, its enabling technologies, main building blocks, and hands-on experience through projects utilizing public cloud infrastructures (Amazon Web Services (AWS) and Microsoft Azure). Cloud computing services are being adopted widely across a variety of organizations and in many domains. Simply, cloud computing is the delivery of computing as a service over a network, whereby distributed resources are rented, rather than owned, by an end user as a utility. Topics may include distributed computing models and technologies, Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), virtualization, security and privacy issues, performance and systems issues, capacity planning, disaster recovery, Cloud OS, federated clouds, challenges in implementing clouds, data centers, hypervisor CPU and memory management, cloud hosted applications, and other advanced and research topics in cloud computing. Course work will include homework assignments, presentations, and a term project that will provide exposure to scientific research in cloud computing. Material may be covered from sources other than the text including research papers.

COURSE OBJECTIVES

In this course we plan to give students an overview of the field of Cloud Computing, and an in-depth study into its enabling technologies and main building blocks. Students will gain hands-on experience solving relevant problems through projects that will utilize existing public cloud tools. It is our objective that students will develop the skills needed to become a practitioner or carry out research projects in this domain. Specifically, the course has the following objectives:

Students will learn

- 1) The fundamental ideas behind Cloud Computing, the evolution of the paradigm, its applicability; benefits, as well as current and future challenges;
- 2) The basic ideas and principles in data center design; cloud management techniques and cloud software deployment considerations;

- 3) different CPU, memory and I/O virtualization techniques that serve in offering software, computation and storage services on the cloud; Software Defined Networks (SDN) and Software Defined Storage (SDS);
- 4) Cloud storage technologies and relevant distributed file systems, NoSQL databases and object storage;
- 5) The variety of programming models and develop working experience in several of them.

LEARNING OUTCOMES

The primary learning outcomes of this course are five-fold. Students will be able to:

- 1) Explain the *core concepts* of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing.
- 2) Apply fundamental concepts in *cloud infrastructures* to understand the tradeoffs in power, efficiency and cost, and then study how to leverage and manage single and multiple datacenters to build and deploy cloud applications that are resilient, elastic and cost-efficient.
- 3) Discuss *system, network and storage virtualization* and outline their role in enabling the cloud computing system model.
- 4) Illustrate the fundamental concepts of *cloud storage* and demonstrate their use in storage systems such as Amazon S3 and HDFS.
- 5) Analyze various *cloud programming models* and apply them to solve problems on the cloud

BASIC CONCEPTS 9hrs

This module will provide a broad overview of cloud computing, its history, technology overview, benefits, risks and the economic motivation for it. Upon completion of this module, students will be able to:

Explain the concept of “cloud computing”.

Briefly recall the recent history of cloud computing, illustrating its motivation and evolution.

List some of the enabling technologies in cloud computing and discuss their significance.

Discuss some of the advantages and disadvantages of the cloud paradigm.

Articulate the economic benefits as well as issues/risks of the cloud paradigm for businesses as well as cloud providers.

Associate the various layers in the cloud building blocks and differentiate cloud service models.

Define SLAs and SLOs and illustrate their importance in Cloud Computing.

Enumerate and explain various threats in cloud security.

List some of the common cloud providers and their associated cloud stacks and recall popular cloud use case scenarios.

CLOUD INFRASTRUCTURE 9hrs

This module will provide a historical overview of data centers, along with design considerations. Students will learn to apply methods to evaluate data centers, cloud management techniques and software deployment considerations. Upon completion of this module, students will be able to:

Describe the evolution of data centers and outline the architecture of a modern data center.

Indicate design considerations and discuss their impact.

Demonstrate the ability to calculate various power requirements of a data center.

Recall challenges and requirements for a cloud-centric data center and how they differ from large, single-entity warehouse-scale computers.

Explain the cloud software stack and the role of each layer within it.

Identify the need for and techniques behind automation and orchestration of resources, as well as key scheduling considerations in the cloud.

Evaluate programming, deployment and failure considerations when programming the cloud.

Understand the implications of building a multi-tier cloud application to achieve resiliency and elasticity, and the latency implications of such applications

Recall various cloud pricing models and their applicability to various business use cases.

Recall and describe cloud management techniques such as middleware, resource provisioning, metering, and orchestration.

Describe and evaluate different cloud software deployment considerations such as scaling strategies, load balancing, fault tolerance, accounting for tail latencies and optimizing for cost.

CLOUD RESOURCE MANAGEMENT 9hrs

Students will learn how virtualization can allow software and hardware images (e.g., virtual machines) to run side-by-side on a single cloud data center while provided security, resource and

failure isolations. They will understand how virtualization enables clouds to offer software, computation, and storage as services as well as attain agility and elasticity properties. We will discuss resource virtualization in detail and present multiple examples from Xen and VMware. Finally, we will present a real use case such as Amazon EC2. After finishing this unit students will be able to:

Identify major reasons for why virtualization is useful, especially on the cloud.

Explain different isolation types such as fault, resource, and security isolations provided by virtualization and utilized by the cloud.

Indicate how system complexity can be managed in terms of levels of abstractions and well-defined interfaces, and their applicability to virtualization and the cloud.

Define resource sharing as provided by virtualization and discuss how it can be offered in space and time via physical and logical partitioning.

Define virtualization and identify different virtual machine types such as process and system virtual machines.

Identify conditions for virtualizing CPUs, recognize the difference between full virtualization and paravirtualization, explain emulation as a major technique for CPU virtualization, and examine virtual CPU scheduling in Xen.

Outline the difference between classical OS virtual memory and system memory virtualization, explain the multiple levels of page mapping as imposed by memory virtualization, define memory over-commitment and illustrate VMWare memory ballooning as a reclamation technique for memory over-committed virtualized systems.

Explain how CPU and I/O devices can communicate with and without virtualization, identify the three main interfaces, system call, device driver and operation level at which I/O virtualization can be carried, and apply I/O virtualization to Xen.

Outline recent developments in software defined networking and software defined storage from the cloud computing perspective.

CLOUD STORAGE 9hrs

This module will provide a broad overview of storage technologies and concepts of cloud storage. It will also provide a detailed study of Amazon S3, EBS and distributed file systems and databases. Students will be able to:

Describe the overall organization of data and storage.

List the various types of data within the data taxonomy and classify different data types within the data taxonomy.

Identify the problems of scale and management in big data. Discuss various storage abstractions.

Compare and contrast different types of file systems and discuss their design considerations.

Compare and contrast Hadoop Distributed File System (HDFS) with Ceph File System (CephFS).

Compare and contrast different types of databases and discuss their design tradeoffs

Discuss the concepts of cloud object storage. Enumerate the different types of block devices used in data storage.

PROGRAMMING MODELS 9hrs

Students will be given an overview on a variety of cloud-based programming models. Students will understand the benefits and limitations of each so that they can assess applicability based on the problem domain. Students will gain working experience in one of these programming models. Upon completion of this module students will be able to:

Explain the fundamental aspects of parallel and distributed programming models.

Recall and contrast different cloud programming models (MapReduce, Spark, GraphLab and Spark Streaming).

Explain the main execution flow, scheduling and fault tolerance concepts in the MapReduce programming model.

Mode of Delivery

Class lectures and notes are designed to achieve the course objectives. You should read the assigned chapters before class, complete assignments on time participate in class.

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

REFERENCES:

Technical papers from major journals and major conferences on computing, networking, cloud computing.

Distributed and Cloud Computing, 1st edition, Morgan Kaufmann, 2011

INTERNET TECHNOLOGY AND WEB DESIGN

COURSE NAME	Internet Technology and Web Designing
COURSE LEVEL	YEAR 2semester 1
COURSE CREDIT	3CU
CONTACT HOURS	45 HOURS
Course code	SWE 2108

Course Description

This course covers ASP.NET development fundamentals. It provides hands on training of server-side technologies, C# (C Sharp) programming language used for ASP.NET and database development utilizing Microsoft SQL Server Desktop Engine (MSDE). Techniques for integrating ASP.NET pages with the advanced client-side (DHTML) technologies are covered.

Course Objectives:

The aims of the course are:

1. To introduce students to the .Net development environment
2. To enable students to apply core ASP.NET technologies to develop Web applications
3. To train students to author server-side ASP.NET code in C#
4. To train students to create Web applications using ADO.NET to interact with SQL Server
5. To enable students to write ASP.NET pages that integrate into traditional DHTML-driven pages.

Learning Outcomes

On successfully completing this course, students shall be able to:

1. Set up a .Net development including environment, including MSDE
2. Apply core ASP.NET technologies to develop Web applications
3. Write server-side ASP.NET code in C#
4. Create web applications using ADO.NET to interact with SQL Server
5. Write ASP.NET pages that integrate into traditional DHTML-driven pages

Course Outline

This course covers:

- | | |
|-------------------------------------|--------|
| 1. ASP.NET development fundamentals | 3hours |
| 2. Server-side technologies | 3hours |

3. Front-End Web Development	7hours
* Separation of Concerns	
* Progressive Enhancement	
* Responsive Web Design	
* Frameworks: Bootstrap, jQuery, AngularJS, React, etc.	
4. Back-End Web Development	7hours
* Three-Tier Architecture	
* Object-Relational Mapping	
* Frameworks: Django, ExpressJS/NodeJS, etc.	
- Routing	
- Template Languages	20hours

Mode of Delivery

Lectures, Case studies, Tutorials, Practicals, Group work

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

1. Build Your Own ASP.NET Website Using C# & VB.NET, by Zak Ruvalcaba, 1st Edition, ISBN:0-9579218-6-1, 2005 22
2. Internet & World Wide Web: How to Program by Paul J. Deite and Harvey M. Deitel, ISBN-10: 0131752421, Prentice Hall; 4 edition , September, 2007
3. ASP.NET Unleashed: unleashed By Stephen Walther, Sams Publishing, 2003 ISBN 067232542X, 9780672325427
4. Internet Programming with VBScript and JavaScript (Web warrior series), by Kate Kalata, Course Technology, 1st edition, December, 2000. ISBN-10: 0619015233

SYSTEMS ANALYSIS AND DESIGN

Course Name SYSTEMS ANALYSIS AND DESIGN

Course Level Year 2 semester I

Course Credit 4CU

Contact hours 45 Hours

Course code SWE 2108

Brief Course Description

The course uses structured systems analysis and design techniques to develop and document effective computer based information systems. It focuses on the tools and techniques used for analysis and design of systems to meet the increasing need for information within organizations.

Course Objectives

The course aims to provide students with understanding and skills on:

- 1 The systems concepts and the system Development Life Cycle
- 2 The role of the System Analyst in the systems development life cycle
- 3 The techniques of requirements elicitation, specification, and analysis of information systems
- 4 Data modeling in information systems development
- 5 Process modeling in information systems development
- 6 Information system development and maintenance

Learning Outcomes

At the end of the course, students shall be able to:

- 1 Explain the systems theoretical concepts and the system Development Life Cycle
- 2 Explain the role of the System Analyst in the systems development life cycle
- 3 Describe and apply the techniques of requirements elicitation, specification, and analysis to information systems development
- 4 Create a data model for an information systems development project
- 5 Create a process model for an information systems development project
- 6 Explain how Information system construction and maintenance is done

Course Outline

The major topics covered include the areas of:

- Systems theory

2 hours

• The systems development life cycle	3 hours
• The role of the systems analyst	1 hour
• Information Systems Requirements Analysis	3 hours
• Process modeling with data flow diagrams	6 hours
• Logic and process specification	6 hours
• Data modeling with entity relationship diagrams	6 hours
• System construction, Implementation and maintenance	3 hours
• Case study and Tutorials (30 hours)	15 hours

Mode of Delivery

Lectures, Case studies, Tutorials, Seminars, Group work

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

- 1 Hoffer J.A; George, J.F; and Valacich J.S. (2009), Essentials of Systems Analysis and Design. Prentice Hall
- 2 Hannagan, T. (2008), Management Concepts and Practices. Prentice Hall
- 3 Hoffer, J.A, George J. F and Valacich ,J. S (2008) Modern Systems Analysis and Design, (2nd Ed) Addison-Wesley
- 4 Whitten J.L, and Bentley L.D, (2007), Systems Analysis and Design Methods. Prentice Hal
- 5 Kendall K.E. and Kendall J.E (2005) Systems Analysis and Design (6th Ed) Pearson Prentice Hall
- 6 Denis A, Wixon B.H, and Tegarden (2005) Systems Analysis and Design with UML Version 2.0.
- 7 Yeates D. (2004), Systems Analysis and Design. Prentice Hall
- 8 Yeates D. and Wakefield T. (2004), Systems Analysis and design, Prentice Hall

POLITICAL ECONOMY OF UGANDA

Course Name: Political Economy of Uganda
Course Level: Year 2 Semester 1
Course Code: DVS 2105
Credit Units: 3CU
Contact Hours: 45

Course Description

This development studies course is designed to provide a background for a detailed analysis of processes of political and social change in Uganda, and their implications for the country's social, political and economic development. The aim of the course is to present and stimulate debate on the ways in which Uganda's historical development has influenced existing institutions, political structures and processes, society and culture, and to emphasize opportunities and potential risks to future development.

Learning objective: this course aims to:

Give an overview of Uganda's political economy from the colonial era through the years of economic reforms.

To teach on the various development plans for the country. There is a study of the various economic policies of the various regimes.

To teach the Political Economy of Pre-Colonial Societies In Uganda

To expose Imperialism and Colonialism With Special Emphasis On Uganda

To show Modernisation, Dependence and Under-Development In Uganda

The Political Economy of De-Colonisation, Neo-Colonialism and Political Struggles

Learning outcomes

By the end of the course, students should be able to

- Comprehend the factors that have delayed Uganda's development.
- Identify specific policies of the various regimes
- Give a general overview of the political economy of Uganda since independence.
- Analyze Uganda's historical and current political and economic perspectives and how these have affected the development of this nation.
- Identify the stem of the development problems and what can be done to alleviate them
- Foster a spirit of nationalism and also ensure that students merge out as good ambassadors of good governance.

Course Outline

- Pre-colonial Ugandan Society-Production, politics, kingdoms **2hrs**
- Colonialism-Colonial economy, colonial regional division of labor, **2hrs**
- Responses to colonialism **2hrs**
- Nationalism and the birth of Political Parties **2hrs**
- Economic Reforms after Independence **2hrs**
- Import Substitution **2hrs**
- Industrial Policies **2hrs**
- Reforms in the 1970s **2hrs**
- Amin's Economic War **2hrs**
- 1980s World Bank Reforms I and II **2hrs**
- Structural Adjustment Programs Before and After 1986 **2hrs**

National Resistance Movement 23hr

- The 10/15 points programme
- Political and Economic Reforms during NRM: 1986-2000- Decentralization, Privatization, Economic Liberalization, The 1995 Constitution
- The Local Government Act: Gender Inequality and Equality Issues (Affirmative Action)
- AGOA
- Rural Development Policies and Strategies
- -Plan for Modernization of Agriculture in Uganda: impact of PMA, Successes and Challenges
- Uganda's role in Regional and African Context
- Peace Keeping missions in Somalia, DRC etc
- Ideology, Patriotism and Nationalism in Uganda

Mode of delivery

Interactive lectures, tutorials, case studies, group work – students based collaborative learning.

Mode of assessment

1. Continuous assessment (30%)
 - a. Tests (20%)
 - b. Presentations – case studies (5%)
 - c. Group work and participation (5%)
2. Final written examination (70%). The final examination will be similar across Faculties, administered on the same day.

Suggested Readings

Mamdani M. (1976). *Politics and Class Formation in Uganda*. London: Monthly Review Press

Hansen, H & Twaddle M. (Eds). (1991). *Changing Uganda: The Dilemmas of Structural Adjustment and Revolutionary Change*. London: Villers Publications

Jorgensen, J. J. (1982). *Uganda: A Modern History*. Croom Helm Ltd

Kabwegyere, Tarsis B.(1995). *The Politics of State Formation and Destruction in Uganda*, Kampala: Fountain Publishers Ltd

Karugire, S. (1996). *Roots of Instability in Uganda*. Kampala: Fountain Publishers

Karugire S. R. (1980). *A Political History in Uganda*. Nairobi and London: Heinemann

KiwanukaSemakula. (1973). *From Colonialism to Independence: A Reappraisal of Colonial Policies & African Reactions 1870 – 1960*, Nairobi: Kenya Literature Bureau

Moncrieffe Joy. (2004). *Uganda's Political Economy: a synthesis of major thought*

Villadsen, Soren and Francis Lubangaed (1996). *Democratic Decentralisation in Uganda: A New Approach to Local Governance*. Kampala: Fountain Publishers, 1996

YEAR TWO SEMESTER TWO

FORMAL SPECIFICATION FOR SOFTWARE ENGINEERING

Course Name	Formal specification for Software Engineering
Course Level	Year 2 semester 2
Course Credit	4CU
Contact hours	60 Hours

Course code **SWE 2201**

Brief Course Description

Review of set theory, the predicate calculus, relations, relational algebra and formal specification concepts; algebraic and model based specifications; the role of formal specifications in software engineering. The Event-B notation, the role of proof obligations and refinement, the LTL and CTL temporal logics, the model checking approach and techniques.

Learning Outcomes

At the completion of this unit students should be able to:

1. Articulate the role and importance of formal modelling and verification;
2. Develop Event-B specifications; apply Rodin to analyse an Event-B specification and verify proof obligations;
3. Distinguish and evaluate the trade-offs in system modelling using Event-B and temporal logics; develop basic specifications and formulate properties in temporal logics;
4. Utilise a model checker to verify such properties.

Course content: topics to be covered include:

1. Review of set theory, **5hrs**
2. the predicate calculus, relations, relational algebra and formal specification concepts; **8hrs**
3. algebraic and model-based specifications. **6hrs**
4. The B notation, schemas, schema calculus, schema types and operations; **6hrs**
5. data and algorithm design; **4hrs**
6. data and operation refinement; **4hrs**
7. proofs of correctness. **5hs**
8. The place of formal methods in software engineering practice/ groups works **22hrs**

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. An Introduction to Formal Methods, by Antoni Diller, 2nd edition, Wiley, (June 1994), ISBN-10: 0471939730
2. Software Domain Model Integration Methodology for Formal Specification by Joel C. Nonnweiler
3. Languages for System Specification: Selected Contributions on UML, SystemC, System Verilog, Mixed-Signal Systems, and Property Specification from FDL'03
4. Software Engineering with OBJ: Algebraic Specification in Action [Book] by Joseph A. Goguen, Grant Malcolm · Springer Categories for Software Engineering by Jose Luiz Fiadeiro · Springer · Hardback
5. The Use of Formal Specification of Software: June 25–27, 1979, Berlin By Berlin Heidelberg ·
6. Software Engineering, Second Edition by Stephen R. Schach · Taylor & Francis
7. Formal Methods and Models for System Design: A System Level Perspective, by Gupta, R., Le Guernic, P. Shukla, S.K. and Talpin, J.P. (Eds.), 2004, ISBN: 978-1-4020-8051-7

DATA COMMUNICATION TECHNOLOGY

Course Name	Data Communication Technology
Course Level	Year 2 Semester 2
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 2202

Brief Course Description

This course introduces concepts of computer networks within the context of the Internet. It covers fundamental principles, elements and protocols of computer networks and data communication.

Course Objectives

The aims of this course are:

- 1 To introduce students to standards, guidelines and technologies in computer and data communication networks
- 2 To impart knowledge and skills relevant for the design, implementation and maintenance of modern communication networks
- 3 To provide a solid basis on the theoretical and practical understanding of data communication in networks

Learning Outcomes

By the end of this course, the student shall be able to:

- 1 Explain the underlying principles of data communication and networks
- 2 Explain the operation of a range computer networking applications such as email, web, and peer-to-peer file-sharing
- 3 Relate the architecture of the Internet to the underlying design principles
- 4 Illustrate the operation and evaluate the performance of common routing protocols, queuing mechanisms, and congestion control mechanisms
- 5 Develop elements of a network such as gateways and routers that conform to IETF standards with acceptable levels of simplification

Course Outline

- | | | |
|---|---|---------|
| 1 | OSI and TCP/IP models | 7 hours |
| 2 | Transmission of Digital Data and transmitting media | 7 hours |
| 3 | Signals, Modulation & Multiplexing | 6 hours |
| 4 | Compression and decompression | 3 hours |

5	Network LAN technologies(fast ethernet,giga ethernet,virtual lan)	
6	Network topologies	
7	Transmission models	
8	Transmission impairments	
9	Wireless networks	3 hours
10	Network Security	3 hours
11	Practical Lessons/group work	20 hours

Mode of Delivery

Lectures, Practicals and Group work

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

1. Andrew S. Tanenbaum, *Computer Network*, Prentice-Hall
2. Douglas E. Comer, *Computer Networks and Internet*
3. <http://www.cisco.com/public/support/tac/documentation.html>
4. <http://www.redhat.com/docs>
5. James F. Kurose and Keith W. Ross. *Computer Networking - A Top Down Approach Featuring the Internet*, 3rd edition, Addison-Wesley, ISBN 0-321-22735-2.
6. *Computer Networks: A Systems Approach*. L. Peterson and B. Davie

SOFTWARE ARCHITECTURE AND DESIGN

COURSE NAME	SOFTWARE ARCHITECTURE AND DESIGN
COURSE LEVEL	Year 2 SEMESTER 2
COURSE CREDIT	3 CU
CONTACT HOURS	45 HOURS
Course code	SWE 2203

Course description

Software architecture defines the primary system qualities of performance, modifiability, and security, into a unifying architectural vision. It is an artifact for early analysis to ensure that a design approach yields an acceptable system. This course also explains the work assignments that must be carried out by design and implementation teams

Course Objectives:

This course aims to enable students to:

1. understand the latest state-of-the-art software architecture
2. understand software architecture system design
3. Understand how system's components are meant to interact with each other.

Course Objectives:

At the end of the course, students shall:

1. Demonstrate familiarity with the latest state-of-the-art software architecture
2. Be able to explain a software system design
3. Explain how system's components interact with each other
4. Describe a software system design for a particular program

Course Content:

The topics to be covered in this course unit are:

- | | |
|---|--------|
| 1. Architectural styles and Patterns | 3hours |
| 2. Components of architectural design | 3hours |
| 3. Connectors, components and composition | 6hours |
| 4. Architectural design guidance and Tools for architectural design | 3hours |
| 5. Achieving quality goals with architectural styles | 3hours |

6. Formal models and specifications	6hours
7. Analyzing software architecture with SAAM	6hours
8. Architecture description languages (ADLs)	6hours
9. Architecture-based development	3hours
10. Patterns in software architecture	3hours
11. Reusing software architecture	3hours

Mode of Delivery

Lectures, Case studies, Group work

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. Software Architecture in Practice, by Len Bass, Paul Clements and Rick Kazman, Addison Wesley 1998.
2. Software Architecture: Perspectives on an Emerging Discipline, by Shaw and Garlan, Prentice Hall 1996.
3. "UML Components: A simple process for specifying component-based software", by Chessman and Daniels, Addison-Wesley 2000.

SOFTWARE METRICS

Course Name
Course Level
Course Credit
Contact hours

SOFTWARE METRICS
Year 2 semester 2
3CU
45 Hours

Course code

SWE 2204

Course Description

This course helps students appreciate the role software metrics play in the control and management of the software development process. Software metrics provide a quantitative basis for development and validation of models of the software development process. Effective management of any process requires quantification and measurement.

Course Objectives

The course aims to provide a student with:

1. An understanding of the need for software metrics and classification of software metrics
2. Knowledge of basic terminology and fundamental facts about software metrics and process models
3. Understanding of the importance of software metrics in the control and management of software development process
4. Knowledge of the fundamentals of measurement theory
5. Understanding of software product metrics , process metrics and measuring management

Learning Outcomes

Upon successful completion of this course students should be able to:

1. Explain basic terminology and state fundamental facts about software metrics and process models
2. Identify essential elements of given metrics describe interrelationships among its various elements and describe circumstances or environment in which its use is appropriate
3. Describe the importance of software metrics in the control and management of the software development process.

4. Understand the foundations of measurement theory and models of software engineering measurement
5. Appreciate software products metrics, software process metrics and measuring management.

Course Content:

The course is composed of the following basic modules:

- | | |
|---|--------|
| 1. Need and definition of Software metrics, | 3hours |
| 2. Classification and measurement of scales for software metrics | 3hours |
| 3. Measurement theory | 4hours |
| 4. Software product and process measurements | 4hours |
| 5. Measuring internal product attributes: size and structure | 4hours |
| 6. Measuring external product attributes: quality, measuring cost and effort, | 4hours |
| 7. Measuring software reliability, software test metrics, object-oriented metrics | 4hours |
| 8. Measurement management | 4hours |
| 9. Tutorials and group work | |
| 15hours | |

Mode of Delivery:

Lectures, Tutorials, Group work

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Software Metrics: A Rigorous and Practical Approach, (2nd ed.) (638p.), N.E. Fenton and S.L.
2. Pfleeger, PWS Publishing, 1998. ISBN 0-534-95425-1. Additional Recommended Text and Reference Books:
3. Metrics and Models in Software Quality Engineering, Stephen H. Kan, 2nd ed. (560 p.), Addison- Wesley Professional (2002). ISBN: 0201729156.

4. Software Engineering Measurement, John C. Munson, Auerbach Publications, 2003 (443 pages) ISBN: 0849315034
5. Software Metrics: Measurement for Software Process Improvement, BA Kitchenham, Blackwell Pub, 1996. ISBN: 1855548208.
6. Applied Software Measurement: Assuring Productivity and Quality, C. Jones, McGraw-Hill, 1996

USER INTERFACE DESIGN AND DEVELOPMENT

Course name: User Interface Design and Development

Course Level: Year Two, Semester Two

Credit Units: 4

Course code: SWE 2205

Description: The course introduces the principles of user interface development, focusing on design implementation and evaluation. Command-driven interfaces using current operating systems, graphic user interfaces using computer aided tools. The development of interfaces using Object-oriented programming languages, scripting languages, simulation software. This course accounts for user-interface development of Websites, Web applications, and traditional client-server applications (and can be extended to mobile platforms as well).

Aims and Objectives: By the end of this course, students will have proper knowledge on:

- Why User interface is crucial in designing
- Usability as regards Graphical and Web design User interface development
- The Interface design life cycle and how prototyping can be used
- Principles of good design
- UML and why it is important in UID
- Proper window selection, interaction devices and testing interfaces
- Use of graphical application tools use: Visual Basic, Macromedia Fireworks, GUI design
- Viewer, Sysbase Power builder, Netbeans IDE
- Small devices interface development: Introduction to WML, J2ME or Android

Course Learning Outcomes: By the end of this course students will be able to:

- understand the issues involved in developing and evaluating interfaces to interactive applications;
- demonstrate an awareness of human-computer interface standards and guidelines;
- apply the principles, concepts and models of user-centered design methods to the development
- and evaluation of interactive system interfaces;
- Apply the concepts of color theory, font terminology, layout and graphic design elements in terms of visual user interface design.

Course Content

- Why User interface matters (6 hours)
 - Defining the user interface
 - Importance of good design
 - Human computer interaction Vs GUI
 - Usability
- Design Techniques (3 hours)
 - Icon design
 - Interaction Styles
 - Layout
- User Interface design Process (6 hours)
 - Phases of UI development
 - UI development activities
 - Assessing your UI development practices
 - Case study of a Web application's UI development
- Prototyping (3 hours)
 - Prototyping techniques
- UML (6 hours)
 - Use cases, Object and class diagrams
 - Activity diagrams, state diagrams,
- Principles of good interface and screen design (3 hours)
 - Human consideration in interface and screen design
- Design Tools for Effective development (3 hours)
 - Computer aided tool like Visual Basic, Macromedia Fireworks, GUI design Viewer
- Mobile devices interface development (6 hours)
 - WM, XML and WAP

(e) Mode of Delivery:

- Lectures
- Individual or group research projects
- Hands-on assignments
- Class Presentations
- Tutorials

(f) Methods of Assessment:

Assessment will be in form of

Continuous Assessment (40%)

Final Examination (60%)

Key Reading List:

- (i) Wilbert, O. G, (2007) The Essential Design to User Interface Design: An Introduction to GUI Design Principles and Techniques (3rd Ed) Wiley
- (ii) Stone, D., Carolline, J., Woordroffe, M., and Minocha, S. (2005) User Interface Design and Evaluation. Morgan Kaufmann

DISTRIBUTED PROCESSING

Course Name	DISTRIBUTED PROCESSING
Course Level	Year 2 semester 2
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 2206

Brief Course Description

Modern computer systems rely increasingly on distributed computing mechanisms, implemented often as clusters, web services, grids and clouds. Distributed computing systems can provide seamless (or web-like) access to a variety of networked resources, e.g. processing cores, large data stores and information repositories, expensive instruments, high-speed links, sensor networks, and multimedia services for a wide range of applications.

This unit provides foundation knowledge and understanding of the basic mechanisms required to implement distributed computing systems, especially clouds, grids, web services and clusters.

Topics covered include:

Introduction to parallel and distributed computing mechanisms, **6hrs** concurrency and synchronisation, **6hrs** monitors, deadlocks, concurrent program analysis - Deadlock, Safety & Liveness properties, **8hrs** Threads processes and IPC (**6hrs**). LVS and Beowulf Clusters. Concurrency control (**4hrs**) Distributed databases and java database connectivity (**4 hrs**). distributes security java security-**5hrs**, **web services 3hrs**, computational and service-oriented grids. **5 hrs** Gridservices, **3hrs**, Clouds and Peer-to-Peer (P2P) networks **4hrs** Distributed applications, and their performance and reliability in relation to processor and network performance constraints. **6 hrs**

Outcomes

At the completion of this unit students will be able to:

1. understand basic problems in distributed computing, especially in relation to concurrency, parallelism, synchronisation, deadlocks, safety and liveness properties;

2. understand differences between various distributed computing models and widely used distributed computing schemes;
3. understand basic functional and performance concepts in grids and clouds and identify frequent causes of performance problems in grid applications;
4. understand basic software and hardware reliability concepts in grids and identify frequent causes of reliability problems in grid applications;
5. discuss some of the enabling technologies e.g. high-speed links, emulators and storage area networks for building computer grids and clouds;

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. S. Tanenbaum and M. V. Steen, Distributed Systems: Principles and Paradigms, Second Edition, Prentice Hall, 2006, ISBN: 0132392275. Reference Books:
2. G. Coulouris, J. Dollimore, and T. Kindberg, Distributed Systems: Concepts and Design, 3rd Edition, Addison-Wesley, 2000, ISBN: 0201619180.
3. R. Anderson, Security Engineering: A Guide to Building Dependable Distributed Systems, John Wiley & Sons, 2001, ISBN: 0471389226.

INDUSTRIAL TRAINING I

Course Name	INDUSTRIAL TRAINING I
Course Level	YEAR 2 SEMESTER 2
Course Credit	4CU
Contact hours	60
Course code	SWE 2207

Course Description

Industrial Placement promotes and develops long-term strategic teaching and research alliances between higher education institutions and industry. It plays a crucial role in promoting and facilitating practical training by providing students the opportunity to apply advanced knowledge and skills acquired in class to real world industry based problems and also gain exposure to real life work experience.

Course Objectives

The course aims to enable students:

- 1 Understand the different types of Computing careers and their roles in a real world business organization.
- 2 Apply and translate computing knowledge and best practice to real industry based problems
- 3 Acquire soft skills to enhance effective participation in group based industry projects
- 4 Learn and practice good working ethics and communication skills with industry based staff and Supervisors.

Learning Outcomes

By the end of this course, the student should be able to:

- 1 Explain the different types of Computing careers and their roles in a real world business organization.
- 2 Demonstrate ability to apply and translate computing knowledge and best practice to industry based problems
- 3 Demonstrate creativity and innovation in solving problems related to real-life projects
- 4 Demonstrate soft skills for enhanced effective participation in group based industry projects

- 5 Demonstrate and practice good working ethics, communicate effectively and exhibit good interpersonal skills with industry based staff and Supervisors.

Course Outline

The major areas that will be emphasized during industrial training include

- | | |
|---|---------|
| 1 Orientation to Industry, Career Prospects & Human Resource Policy | 4hours |
| 2 ICT Infrastructure in Industry | 6hours |
| 3 Communication Network Infrastructure | 4hours |
| 4 Information Management | 10hours |
| 5 Systems Administration | 4hours |
| 6 Website Development | 4hours |
| 7 IT Quality Assurance | 4hours |
| 8 Audit and Review ICT Infrastructure Maintenance Manuals | 4hours |
| 9 Audit and Review IT and Communication Policies | 4hours |
| 10 Audit and Review Information Security | 4 hours |
| 11 Project Report Writting | 12 |
| hours | |

12 Mode of Delivery

Mentorship

Course Assessment

- | | |
|--|-----|
| • Evaluation Report by Industry Based Supervisor | 40% |
| • Interview by Academic Supervisor | 30% |
| • Written Industrial Training Project Report | 30% |

No particular reference will be used for this course unit,except whatever is recommended by the student supervisors.

CITIZENRY, PROFESSIONALISM, GLOBALIZATION AND ENTREPRENEURSHIP

Course Name: Citizenry, Professionalism, Globalization and Entrepreneurship
Course Level: Year 2 Semester 2
Course Code: DVS 2201
Credit Units: 3 CU (45hrs)

Description

Effective citizenship is a very important deciding factor in nation building in both developed and developing countries. Also, national development plans (economic, social, political, scientific and technological aspects) cannot be well implemented without the contribution of effective and incorruptible citizens. This course looks at the present situation in Uganda society and posits that the nation cannot develop or make necessary progress without effect citizens who are incorruptible and committed. One of the challenges facing the public as well as private sector is professionalism. Increasingly, institutions have hired professionals, but this has not improved the quality of service; the quality of service has slowly gone down. The current paradigm of new public management with emphasis on market mechanisms in the delivery of public service comes in as a redress of public failure to deliver service. Within the New Public Management framework the private – public partnership ideals have been introduced in the public service sector. Despite these approaches, service delivery continues to be challenging; there is increased outcry over corruption with increased number of cases reported from time to time. This course unit examines professions and the extent to which professionalism is practiced in Uganda.

This course explores the socioeconomic implications of globalization and some fundamental changes that have taken place in the global socioeconomic system. This course examines the economic, social, political, environmental, and cultural dimensions of globalization from African perspectives. Major global issues such as democratic governance, inequalities of development, poverty eradication, conflict and war, disease control, climate change, financial crises, natural resource exploitation, and diasporization, cultural change, and appropriation will be illustrated and analyzed from the perspective of Africa's experience of globalization.

Aims and objectives

Furthermore the course aims at examining the nature of new business opportunities in Africa, to gain insights on how a new generation of African entrepreneurs is exploiting these opportunities and meeting the increasingly complex entrepreneurial challenges as Africa merges into the mainstream of global commerce.

Learning outcomes

- Identify and describe the major global issues confronting Africa today
- Identify and analyze the economic, political and socio-cultural implications of foreign investment in Africa
- Understand the role of entrepreneurship for Africa's development
- Explain the relevance of professional ethics for society's development
- Apply the principles of ethics in the critical areas of development practice

Course outline

The outline is divided into three major sections

1. Globalization and Development

- Critical and historical approach to Globalization
- Contemporary Globalizing processes (Cash flows, State M-Market Relations, Transnational Corporations, Global Commodity Chains, Inequality and Poverty)
- Labor Migration
- Globalization and Imperialism (Post Cold War Imperial and civil wars, Global and regional Challengers to US Hegemony: Russia and China)
- Globalization Democracy and Culture (Human Rights, Democratization, cosmopolitanism, standardization and homogenization)
- Foreign Aid, Grants/Donations: Implications on Sustainability
- Alternative to Neo-liberal Globalization (Global Labor Movement, Transnational Social Movements, NGOs and Environment)
- New Economic Arrangements: The fall of USSR and the triumph of Capitalism
- Emerging Economies: the BRICS and the rise of China as a super power , The NICs (Asian Tigers)
- Modern Slavery and its Effects- Labor, Human Trafficking, Terrorism and Religious Fundamentalism

2. Citizenry, Professionalism and Development

- Definition and Key concepts: Ethics, Morality, Integrity
 - Theories of ethics- Moral Theories, Deontological Theories, The nature law Theory, Virtue Theory
 - Professional ethics
 - Responsible citizenship
- civil ethics and professional ethics, responsible citizenship (patriotism) –Institutional trust (trust in Public institutions), self awareness, confidence and community participation, corruption and the development agenda.

3. Entrepreneurship and Development in Africa

- Entrepreneurship and management, entrepreneurship skills development, career opportunities in Entrepreneurship, finance and development, basics in financial management, Uganda's policies on entrepreneurship development
- Roots of entrepreneurship: economic, cultural, social considerations
- Settings for entrepreneurship: private and public sectors; large, small, and micro enterprises
- Private enterprise and industrial development-Small Scale, medium and large scale, industries
- The role of Foreign Direct investments
- Entrepreneurship in comparative perspective: The nature and diversity of entrepreneurship and entrepreneurs in Africa: Emerging business opportunities
- Competencies and skills are necessary to succeed in an African business context
- Challenges and innovation in business spheres
- The role of leadership in entrepreneurship

Mode of delivery

Interactive lectures, tutorials, case studies, group work – students based collaborative learning.

Mode of assessment

1. Continuous assessment (30%)
 - a. Tests (20%)
 - b. Presentations – case studies (5%)
 - c. Group work and participation (5%)
2. Final written examination (70%). The final examination will be similar across Faculties, administered on the same day.

Suggested Readings

Aman, K. (ed) 1991 Ethical Principles for Development: Needs, Capacities or Rights? Upper Montclair, NJ: Institute for critical Thinking.

Elkan, W. (1988). "Entrepreneurs and Entrepreneurship in Africa." The World Bank Research Observer 3(2): 171-188.

CS, Z., S. Desai, et al. (2008). "Entrepreneurship, economic Development and Institutions." Small Business Economics 31(3): 219-234.

Appiah, Francis, Chimanikire Donald and Thorvald Gran (2004) Introduction- Professionalization and Professionalism in Appiah, Francis, Chimanikire Donald and Gran Thorvald (2004), *Professionalism and Good Governance in Africa*, Oslo, pp 18- 20

YEAR THREE SEMESTER ONE

HUMAN COMPUTER INTERACTION

Course Name	HUMAN COMPUTER INTERACTION
Course Level	Year 3 Semester I
Course Credit	3CU
Contact hours	45 Hours
Course code	SWE 3101

Course Description

The course helps students to appreciate the delicate and divergent nature of human and computer interactions. It covers the dynamics of user interface designs including their evaluations as well as usability issues for both software and related devices.

Course Objectives

The aims of this course are equip students with knowledge and skills:

- 1 To appreciate established human-computer interaction approaches and techniques.
- 2 To design user interfaces
- 3 To use appropriate software evaluation techniques and heuristics

Learning Outcomes

On completion of this course, the student shall be able to:

- 1 Apply established human-computer interaction approaches and techniques
- 2 Produce conceptual and physical designs using prototyping methods
- 3 Evaluate software interfaces using appropriate heuristics

Course Outline

- | | |
|--|-----------------|
| 1 Human Computer Interaction concepts | 5 hours |
| 2 User Interfaces | 5 hours |
| 3 Design principles of HCI | 5 hours |
| 4 Evaluation paradigms, frameworks, heuristics, usability testing | 5 hours |
| 5 Software posture, flow, navigation, Interaction, language and attributes | 5 hours |
| 6 Human and social factors | 5 hours |
| 7 Practicals/ Tutorials | 15 hours |

Mode of Delivery

Lectures, Practicals/ Tutorials

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

- Human computer interaction by Alan Dix, Janet Finlay, Gregory D. Abowd, Russell Beale, Prentice-Hall Europe 1993, 1998, ISBN-13: 978-0-13-046109-4
- Interaction design : beyond human- computer interaction1 Jennifer Preece, Yvonne Rogers, Helen, Copyright O **2002** John Wiley & Sons, Inc., ISBN **0-471-49278-7**

SOFTWARE RELIABILITY & TESTING

COURSE NAME	SOFTWARE RELIABILITY & TESTING
COURSE LEVEL	YEAR 3 semester 1
COURSE CREDIT	4CU
CONTACT HOURS	60 HOURS

Course code **SWE 3102**

Course description

This course presents an overview of the Software Reliability process, techniques and models. Software Reliability is the probability of failure-free software operation for a specified period of time in a specified environment and is an important factor affecting system reliability. The high complexity of software is the major contributing factor of Software Reliability problems.

Course Objectives

The course aims to provide students with knowledge and skills in:

1. The various software reliability process growth models
2. The various metrics used for software reliability measurement
3. The techniques that can be used to improve and predict software reliability
4. The various techniques applied to software testing
5. The application of a software reliability process model to a given case scenario
6. The software reliability process

Learning Outcomes

Upon successful completion of this course the student shall be able to:

1. Explain the software reliability process
2. Describe the various software reliability process growth models
3. Describe and explain the various metrics used for software reliability measurement
4. Explain the techniques that can be used to improve and predict software reliability
5. Describe and Explain the various techniques applied to software testing
6. Apply a software reliability process model to a given case scenario

Course content

This course covers the following topics:

- | | |
|---|---------|
| 1. The software testing and reliability concepts | 4 hours |
| 2. The software reliability process and its application | 4 hours |

3. Factors influencing software reliability	4 hours
4. Software reliability Engineering techniques	4 hours
5. Software reliability models and Measurement	6 hours
6. software reliability improvement and prediction Techniques	6 hours
7. Metrics for software reliability prediction	6 hours
8. Soft ware System testing	8 hours
9. Practicals and group work	18 hours

Mode of Delivery

Lectures, Practicals and Group work

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. Software Reliability Engineering: More Reliable Software Faster and Cheaper, John D. Musa, (632p.), Authorhouse, 2nd edition, 2004.ISBN 1418493872.
2. Handbook of Software Reliability Engineering, Michael R. Lyu (Editor), McGraw Hill (1996).ISBN:0-07-039400-8.
3. Software Reliability: Measurement, Prediction and Application, J.D. Musa, A. Iannini, K. Okumoto,(621 p.), McGraw-Hill (1987).ISBN 0-07-044093-X.
4. Effective Methods for Software Testing, William E. Perry, 2nd edition, John Wiley and Sons (2000).ISBN: 0-471-35418-X.

SOFTWARE PROJECT MANAGEMENT

Course Name	SOFTWARE ENGINEERING PROJECT MANAGEMENT
Course Level	Year 3 semester 1
Course Credit	3CU
Contact hours	45

Course code **SWE 3103**

Course Description

The module defines the principles of implementing successful projects. Various aspects are considered including Costing, budgeting, risk management, human resource change management and scheduling as critical factors to the success of projects.

Course Objectives

The Course aims to provide students with understanding and skills on:

1. Project Planning, scheduling and Control
2. Time, Cost and Quality management
3. Building and leading multi-disciplinary and effective teams
4. Risk and change management
5. project monitoring using appropriate project management software

Learning Outcomes

At the end of this course, the student shall be able to:

1. Explain project management concepts and roles through all projects
2. Manage a project to its satisfactory conclusion on time and within budget
3. Integrate the concepts of risk and change management in project management
4. Demonstrate the ability to use project monitoring tools and techniques to ensure successful project implementation

Course Outline

The major areas that will be covered include:

- | | |
|--|---------|
| 1. Overview of Project Genesis | 5 hours |
| 2. Project Planning, scheduling and control | 5 hours |
| 3. Estimating, Cost and Budgeting | 5 hours |
| 4. Project Risk Analysis and Management | 5 hours |
| 5. Project stakeholder analysis and Project Team Structuring | 6 hours |

6. Monitoring Progress – Processes tools and Techniques	5 hours
7. Configuration Management	5 hours
8. Quality management	4 hours
9. Practicals	5 hours

Mode of Delivery

Lecturers and Tutorials

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

- Software project management by Bob Hughes and MK, 5th edition. McgralHill

TECHNICAL DOCUMENTATION FOR SOFTWARE ENGINEERS

Course Name	Technical Documentation for Software Engineers
Course Level	Year 3 semester 1
Course Credit	4CU
Contact hours	60 Hours
Course code	SWE 3104

Brief Course Description

This unit covers problems with paper-based and on-line documentation; types of technical documentation used in software engineering; the role of various different forms of technical documentation in the software development process; document specifications; the context of technical writing; the writing process (analysis, planning, generation, testing, revision and maintenance of written texts); document publication techniques (such SGML, LaTeX, and/or XML); the role of hypertext, hypermedia and markup languages in technical documentation; small-volume and large-volume hypertext; collaborative hypertext; intelligent hypertext.

Outcomes

On successful completion of this unit, students should be able to:

1. Produce well-organised and clear technical documentation;
2. Explain the purposes and uses of the different types of technical documentation, including code documentation, internal design documentation, external design documentation, reference manuals, guides, and tutorials;
3. Distinguish between the different types of tools for producing documentation (text editors, formatters, typesetters, desktop publishers, graphics tools, printing and viewing tools) and select tools that are appropriate for specified applications;
4. Produce technical documentation that is written in an appropriate style and at an appropriate level for different classes of readers, including fellow software engineering professionals, managers, clients, and end-users;
5. Evaluate the correctness, appropriateness, and usability of written documentation, and apply strategies to improve these;
6. Correctly use specified software tools to create and publish technical documentation.

Course content

1. introduction to Topic, Overview of course and relevance to traineeship and workplace; technical documentation in various applications and with products; purpose; types of documentation understanding the context of writing **8 hrs**
2. Types of documentation; Copyright; document standards - industry and organisation; document control and revision, storage and revision. Client requirements, scope. **6hrs**
3. Designing documentation; Design tools; systems requirements; flow charts; style documents and templates; format of documentation Single-source document production **6hrs**
4. Develop Documentation: Document Structure, Style documents; templates; design and mapping of documents; use of images and graphics **7 hrs**
5. Evaluate and Edit Document; Digital document design; Using plain English; Editing; proofreading; Contents and indexes **4 hrs**
6. Evaluate whether documents meets client's needs and prepare for publication; **5 hrs**
7. Writing for technical and nontechnical audiences **4hrs**
8. Working with technical writers **4 hrs**
9. Practical projects to be done in group to learn team work **6 hrs**

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Technical Documentation and Process Jerry C. Whitaker, Robert K. Mancini
2. Documenting Software Architectures: Views and Beyond (2nd Edition) Addison-Wesley
3. The SketchUp Workflow for Architecture by Michael Brightman · Wiley
4. User-centered Information Design for Improved Software Usability by Pradeep Henry ·
5. The Art of Software Architecture: Design Methods and Techniques [Book] by Stephen T. Albin · Wiley

MODELLING AND SIMULATION

Course name: Modelling and Simulation

Course Level: Year Two, Semester 1

Credit Units : 4

Course code: SWE 3105

(a) Description: Modelling is the fundamental process to understand, design and communicate about systems and its related components. Every object of interest in computer technology is a product of human thought, and is, therefore, amenable to analysis and design as a system; modelling is the framework in which this activity is formalized with a very specific goal in mind. In this course, we outline some of the standard modelling tools we require to design and analyse a number of system models. Simulation is a way to check the performance of a system without actually building it; and therefore, it does provide a rational basis for system design architecture in a substantial way. In this course, we indicate usage of simulation techniques in approaching both discrete and continuous systems, respectively. Knowledge of any structured programming language would suffice for most cases.

(b) Aims and Objectives: The aim of this course is to introduce students to basic simulation methods and tools for modelling and simulation of continuous, discrete and combined systems.

(c) Course Learning Outcomes: Upon completion of this course, the students should be able to:

- Explain the concept of modelling and the use of abstraction that allows the use of a machine to solve a problem.
- Explain and give examples of the benefits of simulation and modelling in a range of important application areas.
- Describe the relationship between modelling and simulation, i.e., thinking of simulation as dynamic modelling.
- Create a simple, formal mathematical model of a real-world situation and use that model in a simulation.
- Explain the difference between validation and verification of a model; demonstrate the difference with specific examples.
- Differentiate among the different types of simulations, including physical simulations, human guided simulations, and virtual reality.
- Demonstrate the ability to apply the techniques of modelling and simulation to a range of problem areas.
- Choose an appropriate modelling approach for a given problem or situation.
- Describe several approaches to validating models.

- Create a simple display of the results of a simulation.

(d) Course Content

- Introduction to Modelling and Simulation (3 hours)
 - Models as abstractions of situations
 - Simulations as dynamic modelling
 - Simulation techniques and tools, such as physical simulations and human-in-the-loop guided simulations.
 - Foundational approaches to validating models
- Introduction to Processes, Systems, Models, and Simulation (3 hours)
- Purpose of modelling and simulation including optimization; supporting decision making, forecasting, safety considerations; for training and education. (3 hours)
- Trade-offs including performance, accuracy, validity, and complexity. (3 hours)
- The simulation process; identification of key characteristics or behaviors, simplifying assumptions; validation of outcomes. (3 hours)
- Model building: use of mathematical formulas or equations, graphs, constraints; methodologies and techniques; use of time stepping for dynamic systems. (3 hours)
- Formal models and modelling techniques: mathematical descriptions involving simplifying assumptions and avoiding detail. (12 hours)
 - Examples of techniques include:
 - Monte Carlo methods, Stochastic processes
 - Queuing theory
 - Petri nets and colored Petri nets
 - Graph structures such as directed graphs, trees, networks
 - Games, game theory, the modelling of things using game theory
 - Linear programming and its extensions
 - Dynamic programming
 - Differential equations: ODE, PDE
 - Non-linear techniques
 - State spaces and transitions
 - Assessing and evaluating models and simulations in a variety of contexts; verification and validation of models and simulations. (6 hours)
 - Important application areas including health care and diagnostics, economics and finance, city and urban planning, science, and engineering. (3 hours)
 - Software in support of simulation and modelling; packages, languages. (3 hours)

(e) Mode of Delivery:

- Lecture Method

- Tutorials
- Class Presentations

(f) Methods of Assessment:

Assessment will be in form of

Continuous Assessment (40%)

Final Examination (60%)

(g) Key Reading List:

(i) Law, A.M. and Kelton, D. W. (1991) Simulation Modelling & Analysis. McGraw Hill

(ii) Sokolowski, J. A and Banks, C. M. (2009) Principles of Modelling and Simulation: A Multidisciplinary Approach. Wiley

(iii) Rossetti, M. D. (2009) Simulation Modelling and Arena. Wiley

(iv) Shiflet, A. B., and Shiflet, G.W. (2006) Introduction to Computational Science: Modelling and Simulation. Princeton University Press

DATA MINING

Course name: Data Mining

Course Level: Year Three, Semester 1

Credit Units : 4

Couse code: SWE 3106

Description: This course introduces students to a comprehensive coverage of well-known data mining topics including classification, clustering and association rules. A number of specific algorithms and techniques under each category will be discussed. Methods for feature selection; dimensionality reduction and performance evaluation will also be covered. Students will learn and work with appropriate software tools and packages used in data mining.

Aims and Objectives:

- The main objectives of this course are to provide an:
- Understanding of the value of data mining in solving real-world problems.
- Understanding of foundational concepts underlying data mining.
- Understanding of algorithms commonly used in data mining tools.
- Ability to apply data mining tools to real-world problems.

Course Learning Outcomes: Upon completion of this course, the students should be able to:

- Display a comprehensive understanding of different data mining tasks and the algorithms most appropriate for addressing them.
- Evaluate models/algorithms with respect to their accuracy.
- Demonstrate capacity to perform practical work that requires the application of data mining techniques.
- Critique the results of a data mining exercise.
- Develop hypotheses based on the analysis of the results obtained and test them.
- Conceptualize a data mining solution to a practical problem.

Course Content

- What Motivated Data Mining? Why Is It Important? **(4 hours)**
 - What Is Data Mining?
 - Data Mining-On What Kind of Data?
 - Relational Databases, Data Warehouses, Transactional Databases.
- Advanced Data and Information Systems and Advanced Applications
- Data Mining functionality-What Kinds of Patterns Can Be Mined?
- Data Preprocessing (4 hours)
 - Why Preprocess the Data?
 - Data Cleaning: Missing Values, Noisy Data, Data Cleaning as a Process.
 - Data Integration and Transformation: Data Integration, Data Transformation.

- Data Reduction: Data Cube Aggregation, Attribute Subset Selection, Dimensionality Reduction, Numerosity Reduction.
- Data Discretization and Concept Hierarchy Generation: Discretization and Concept Hierarchy Generation for Numerical Data, Concept Hierarchy Generation for Categorical Data.
- Mining Frequent Patterns, Associations, and Correlations (8 hours)
 - Market Basket Analysis: A Motivating Example, Frequent Itemsets, Closed Itemsets, and Association Rules, Frequent Pattern Mining.

Efficient and Scalable Frequent Itemset Mining Methods:

- The Apriori Algorithm: Finding Frequent Itemsets Using Candidate Generation, Generating Association Rules from Frequent Itemsets, Improving the Efficiency of Apriori, Mining Frequent Itemsets without Candidate Generation, Mining Frequent Itemsets Using Vertical Data Format, Mining Closed Frequent Itemsets.
- Classification and Prediction (**8 hours**)
 - What Is Classification? What Is Prediction?
 - Issues Regarding Classification and Prediction: Preparing the Data for Classification and Prediction, Comparing Classification and Prediction Methods.
 - Classification by Decision Tree Induction, Decision Tree Induction, Attribute Selection Measures, Tree Pruning, Scalability and Decision Tree Induction.
 - Bayesian Classification: Bayes' Theorem, Naive Bayesian Classification, Bayesian Belief Networks.
 - Rule-Based Classification: Using IF-THEN Rules for Classification.
 - Rule Extraction from a Decision, Rule Induction Using a Sequential Covering Algorithm.
 - Other Classification Methods: Genetic Algorithms, Rough Set Approach, Fuzzy Set Approaches.
 - Prediction: Linear and Nonlinear Regression.
 - Accuracy and Error Measures: Classifier Accuracy Measures, Predictor Error Measures.
- Cluster Analysis (8 hours)
 - Types of Data in Cluster Analysis: Interval-Scaled Variables, Binary Variables, Categorical,
 - Ordinal, and Ratio-Scaled Variables, Variables of Mixed Types, Vector Objects
 - A Categorization of Major Clustering Methods
 - Partitioning Methods, Classical Partitioning Methods: k-Means and k-Medoids.

Mode of Delivery:

- The course consists of a lecture component and a project component.
- The lecture component introduces the basic concepts and techniques of logic programming.

- The project component will be one or several projects related to logic programming.

Methods of Assessment:

Assessment will be in form of

Continuous Assessment (40%)

Final Examination (60%)

Key Reading List:

(i) Hand, D., Mannila, H. and Smyth, P. (2001) Principles of Data Mining MIT Press

(ii) Tan, P., Steinbach, M., and Kumar, V. (2006). Introduction to Data Mining Pearson Education (Addison Wesley)

(iii) Witten, I., Frank, E., and Hall, M. (2011) Data Mining-Practical Machine Learning Tools and Techniques with Java Implementations. (3rd Edition). Morgan Kaufmann

(iv) Han, J., Kamber, M. and Kaufmann, M. (2006) Data Mining Concepts and Techniques. Morgan Kaufmann

MOBILE NETWORKS AND COMPUTING

Course Name	Mobile networks and Computing
Course Level	Year 3 semester 1
Course Credit	4CU
Contact hours	60 Hours

Course code **SWE 3107**

Course description

Course Description: This course examines principles, design, implementation, and performance of mobile computing and wireless networking. The aim is to lay a foundation in the student's understanding and skills in mobile computing and wireless networking standards, technologies, application and services. Ideally the course is an integration of Wireless Networking and Mobile Computing. Subjects of study under Mobile Networking will include; Wireless Network technologies (including GSM/GPRS/3G & Wireless LANs), Convergence networks, NextGen, Mobile IP, wireless ATM, Wireless Ad Hoc Networks and Bluetooth. While subjects of discussion under Mobile Computing will include; Mobile Computing Architectures (including SMS/SIM, WAP, I-mode and J2ME/J2EE), mobile OS (e.g. Android) and Mobile Application Development including location-based services

Course Objectives At the end of the course, students will be able to:

- Understand the basic principle, architecture and challenges of wireless networks and mobile computing systems; and
- Demonstrate knowledge of software development related to mobile computing systems
- To introduce students to the theory and practice of Mobile networking and computing.
- To facilitate the development of technical skills in mobile application development platforms particularly J2ME and J2EE.
- To enhance students' skills in mobile application development using Java and other technologies
- To introduce students to the fundamental concepts in wireless technology and mobile computing including standards, technologies, devices and services
- To use and experiment with new technology and cutting-edge projects
- To understand how networking research is done
- Investigate novel ideas in the area via semester-long skill development projects

Course Content

The topics for wireless networking include:

- Introduction to mobile computing and wireless networking principles, As well as mobile computing models in general. **5 hrs**
- The Blue-Tooth and Zigbee short range wireless technologies **4hrs**
- The Wi-Fi Technologies, Wireless LANs, **6hrs**
- Introduction to GSM/GPRS/3G/3GPP/ the SMS Bluetooth **10hrs**
- The Mobile IP technology **3hrs**
- Wireless ATM technology **4hrs**
- Introduction to Mobile application development **10hrs**
- Introduction to J2ME and J2EE platforms **8hrs**
- understanding mobile computing architectures (I-mode, SMS and WAP) **4hrs**
- Introduction to GSM gateways e.g. Kannel **6hrs**

Modes of assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

References

D. Jochen Schiller, “Mobile Communications”, Addison-Wesley, 2000.

- Mark Beaulieu, “Wireless Internet Applications & Architectures: Building Professional Wireless Applications Worldwide”, Addison-Wesley, 2002.
- Qusay H. Mahmoud, “Learning – Wireless Java”, O’Reilly, 2002.
- Introduction to Mobile Computing in Mobile Computing, edited by T. Imielinski and H. Korth, Kluwer, Academic Publishers, 1996.
- John W. Muchow, “Core J2ME, technology & MIDP”, Prentice-Hall, 2002.
- Subrahmanyam Allamaraju, “Professional Java Server Programming J2EE 1.3 Edition”, Wrox Press Ltd, 2001.
- Steve Mann, “Programming Applications with Wireless Application Protocol”, John Wiley & Sons, Inc., 2000.

- Scott B. Guthery and Mary J. Cronin, “Mobile Application Development with SMS and the SIM toolkit”, McGraw-Hill TELECOM, 2000.
- Neil Rbodes & Julie McKeehan, “Palm Programming: The Developer’s Guide”, O’Reilly, 1999.
- Mark Hapner, “Java Message Service API Tutorial and Reference – Messaging for the J2EE Platform”, Sun Microsystems, 2002.

YEAR THREE SEMESTER TWO

EMBEDDED SYSTEMS SOFTWARE

Course Name
Course Level
Course Credit
Contact hours

EMBEDDED SYSTEMS
Year 3 semester 2
4CU
60 Hours

Course code

SWE 3201

Course Description

This course presents assembly language programming as the bridge between high-level programming languages and embedded systems hardware with particular reference to the embedded processor.

Course Objectives

The course aims at:

1. Introducing students to the role of assembly language programming
2. Equipping students with skills required to apply a modular approach to assembly language programming with code reuse
3. Provide students with knowledge of the instruction set of a typical embedded processor
4. Imparting an understanding of memory addressing, the various addressing modes, hardware interrupts and how to use them
5. Enabling students to integrate assembly language subroutines into a high-level language program

Learning Outcomes

Upon successful completion of this course, the student will:

1. Understand the role of assembly language programming
2. Understand the instruction set of a typical embedded processor
3. Be able to employ a modular approach to assembly language programming with code reuse
4. Be able to use embedded systems development tools
5. Understand memory addressing and use various addressing modes
6. Understand hardware interrupts and be able to use them
7. Be able to integrate assembly language subroutines into a high-level language program

Course Outline

Topics covered include:

- | | |
|--|----------|
| 1. An introduction to assembly language | 3 hours |
| 2. The addressing modes, register file, and instruction set of a microcontroller | 12 hours |
| 3. Timers, handshaking, input and output, and analog to digital conversion | 12 hours |
| 4. Interrupts | 8 hours |
| 5. Software control of hardware | 10 hours |
| 6. Practical lessons (30 hours) | 15 hours |

Mode of Delivery

Lectures and Practicals

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. M. Ben-Ari, Principles of Concurrent and Distributed Programming, Prentice Hall, 1990
2. Stuart Bennett. Real-Time Computer Control: An Introduction, Printice Hall International, Series in Systems and Control Engineering, 1988
3. Alan Burns and AndyWellings, Real-Time Systems and Programming Languages, AddisonWesley,third edition, 2001

MOBILE COMPUTING AND APPLICATIONS

Course Name	MOBILE COMPUTING AND APPLICATIONS
Course Level	Year 3 semester 2
Course Credit	4 CU
Contact hours	60 Hours
Course code	SWE 3202

Course Description

This course introduces hardware and software concepts of mobile computing. It then discusses mobile application development technologies.

Course Objectives

The aims of this course are:

1. To introduce students to hardware and software components of mobile computing
2. To equip students with an understanding of mobile operating systems as well as mobile multimedia messaging and web services
3. To impart knowledge and skills required to design and build professional mobile applications

Learning Outcomes

Upon successful completion of the course, students should be able to:

1. Explain the fundamentals of mobile computing including both the hardware and software components
2. Understand the mobile IP stack and mobile web access, technologies and services.
3. Demonstrate knowledge of mobile software architecture, operating system platforms and software development tools for mobile software, content and service development.

Course Outline

The topics to be covered include:

- | | |
|---|---------|
| 1. Fundamentals of Mobile Terminal Hardware | 2 hours |
| 2. Radio, DSP, Memory and CPU components | 3 hours |
| 3. Base Station side radio interface standards | 2 hours |
| 4. The Mobile IP stack and mobile web-browsing | 2 hours |
| 5. The WAP-protocol & Location Information | 3 hours |
| 6. Principles of Multimedia Messaging (SMS, MMS) and web services | 4 hours |

7. Introduction to Mobile Operating System Platforms (OSPs)	6 hours
8. Mobile Applications User Interface	2 hours
9. Software set-up in modern Mobile Terminals	6 hours
10. Fundamentals of Mobile Content	6 hours
11. Mobile web content design	8 hours
12. The .mobi top level domain (TLD) initiative	1 hour
13. Practical Lessons	15 hours

Mode of Delivery

Lectures and Practicals

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. D. Milojicic, F. Douglass and R. Wheeler, editors, Mobility: processes, computers, and agents. Addison Wesley, 1999.

SOFTWARE ENGINEERING INDUSTRIAL MINI PROJECT 1

Course Name	SOFTWARE ENGINEERING Industrial MINI PROJECT 1
Course Level	Year 3 semester 2
Course Credit	4CU
Contact hours	60
Course code	SWE 3203

Course Description

The course introduces students to the integrated team approach to undertaking software Engineering projects from the point of requirements analysis to deploying a tested and well engineered solution, with the assistance of a Project Supervisor (s). It also plays a crucial role in exposing students to building highly effective teams which is very important in “real life” industrial experience.

Course Objectives

The course aims to provide students with understanding and skills in:

1. Project management concepts and roles
2. Planning and controlling projects
3. Considerations for commercialization of software products
4. Software Requirements Analysis, specification and Design
5. Choosing appropriate Software Engineering Processes, Tools and Methods
6. Professional Software construction
7. Software maintenance and Management
8. Software documentation
9. Professional Software quality assurance and Testing

Learning Outcomes

By the end of this course, the student should be able to:

- 1 Explain the nature of Software Engineering projects and their management
- 2 Demonstrate understanding of Finance Planning, Budgeting and Costing in Software Engineering Projects
- 3 Integrate the concepts of Risk Management, Human resource management, control of time and deadlines through application of tools and techniques

- 4 Demonstrate the ability to develop a business plan as part of the Project Proposal for their choice of research project.
- 5 Demonstrate creativity and innovation in the developing software solutions
- 6 Develop the requirements and design specifications
- 7 Attest and practice high-quality Project Management skills in enhancing individual effectiveness and productivity
- 8 Develop Software using state of the art technologies
- 9 Document the software in accordance to well known practices

Course Outline

The major areas that will be covered include:

1	Project Proposal and Business Plans	6 Hours.
2	Requirements Analysis, Specifications and Design	8 Hours.
3	Project Management	4 Hours
4	Software Methodology, Tools and Techniques	6 Hours
5	Software Construction	20 Hours
6	Software Quality Assurance and Testing	10 Hours
7	Project Report Write Up	6 Hours

Mode of Delivery

Presentations, Project supervision

Course Assessment

- Project Proposal 10%
- Project Demonstration 50%
- Project Report 40%

No particular reference will be used for this course unit, except whatever is recommended by the student supervisors.

RESEARCH METHODS IN COMPUTING

Course Name	RESEARCH METHODS IN COMPUTING
Course Level	Year 3 semester 2
Course Credit	3CU
Contact hours	45

Course code **SWE 3204**

Course Description

This course provides students with the theoretical and practical concepts of research. It provides the skills, methods and competences that are necessary to collect, analyse, summarise and present quantitative and qualitative data.

Course Objectives

The aims of the course are to:

1. Provide students with knowledge of conducting research that achieves pre-determined objectives.
2. Describe the various data analysis and reporting techniques.

Learning Outcomes

On completion of this course, the student should be able to:

1. Demonstrate knowledge of how to conduct research that achieves pre-determined objectives.
2. Use the various data analysis and reporting techniques.

Course Outline

- | | |
|---|-----------------|
| 1. Introduction to research methods and Research Approaches | 6 hours |
| 2. Problem formulation | 3 hours |
| 3. Data collection and Data Analysis | 9 hours |
| 4. Data presentation and report writing | 3 hours |
| 5. Research ethical issues | 3 hours |
| 6. Tutorials | 15 hours |

Mode of Delivery

Lectures and Tutorials

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Essentials of Research Methods, by Janet M Ruane, Montclair State University, Jul 2004, ISBN10: 0631230483

COMPILER CONSTRUCTION

Course Name
Course Level
Course Credit
Contact hours

COMPILER CONSTRUCTION

Year 3 Semester 2

4CU

60 Hours

Course code

SWE 3205

DISCRIPTION

This course introduces the student to the theoretical and practical knowledge of compiler construction. Topics covered include: Introduction to compilers, interpreters, and the tools for parsing and translation. Lexical analysis, context free grammars and software tools for their recognition. Attribute grammars and their applications in translation and compiling.

EXPECTED LEARNING OUTCOMES

By the end of the course unit, students should be able to;

- Describe programs related to Compiler
- Discuss the functional decomposition and architecture of a compiler
- Explain the general working process of a compiler for a toy language
- Design and Implementation of a Scanner

COURSE CONTENT

Introduction to Compiler

4hrs

- Programming Languages
- Compiler and Interpreter
- Programs related to Compiler
- Design and Implementation of a Compiler
- Testing and Maintaining of a Compiler
- A Toy Compiler
- Functional Decomposition and Architecture of a Compiler

- General Working Process of a Compiler for a Toy Language

Scanning

4hrs

- Overview
- General Function of a Scanner
- Some Issues about Scanning
- Finite Automata

Definition and Implementation of DFA 5hrs

- Non-Determinate Finite Automata
- Transforming NFA into DFA
- Minimizing DFA
- Regular Expressions
- Definition of Regular Expressions
- Regular Definition
- From Regular Expression to DFA
- Design and Implementation of a Scanner
- Developing a Scanner from DFA
- A Scanner Generator – Lex

Context-free Grammars and Parsing 4hrs

- The Parsing Process
- Context-free Grammars
- Parse Trees and Abstract Syntax Tree
- Ambiguous
- Syntax of Sample Language

Top-down Parsing 4hrs

- Overview of Top-down Parsing

- Three Important Sets
- Left Recursion Removal and Left Factoring 4.3 Recursive-Descent Parsing (basic method)
- Parsing
- Grammar
- Parsing Table
- Parsing Engine
- Parser Generator -- LLGen

Bottom-up Parsing 4hrs

- Overview of Bottom-up Parsing
- Finite Automata for LR(0) Parsing
- Parser Generator

Semantic Analysis 4hrs

- Overview of Semantic Analysis
- Symbol Table
- Semantic Analysis of Types
- Semantic Analysis of Declaration
- Semantic Analysis of Body
- Attribute Grammar and Action Grammar

Intermediate Code Generation 4hrs

- Intermediate Representations
- Intermediate Code Generation of Expressions
- Intermediate Code Generation of Atomic Statements
- Intermediate Code Generation of Structural Statements

Intermediate Code Optimization 4hrs

- Overview of Optimization Techniques

- Basic Block
- Local Optimization of Constant Expressions
- Local Optimization of Common Expressions
- Loop Invariant Expressions
- Global Optimizations

Runtime Environment 4hrs

- Storage Organization and Allocation
- Activation Record and Stack
- Variable Access Environment

Target Code Generation 4hrs

- Overview of Target Code Generation
- Virtual Machine
- Temporary Variables
- Register Allocation
- Translation from Quadruples to Target Codes
- From AST to Target Codes

MODE OF DELIVERY

- Lectures
- Reading assignments
- Hands on in the lab

INSTRUCTIONAL MATERIALS/EQUIPMENT

- Flip Charts
- LCD Projectors
- CDs, DVDs

COURSE ASSESSMENT

- Continuous assessment tests 40%
- End of semester examination 60%

Total 100%

READING MATERIALS

Aho, Alfred V., (1986). Ravi Sethi, and techniques and Tools. Jeffrey Ullman. Compilers: Principles, A: Addison-Wesley.

Appel, Andrew W. (1997). Modern Compiler Implementation in Java. Cambridge, Cambridge University Press, ISBN: 0521583888.

Kenneth C. Loudon. (1997). Compiler Construction: Principles and Practice. PWS Publishing Company, IBSN 0-534-3972-4.

SYSTEMS PROGRAMMING

Course Name
Course Level
Course Credit
Contact hours

SYSTEMS PROGRAMMING-Linux
Year 3 semester 2
4CU
60 Hours

Course code

SWE 3206

Course Description

The course trains learners on how to develop codes and algorithms that can evaluate and improve on the performance of an operating system. It requires prerequisite knowledge in programming and operating systems.

Course Objectives

The aims of the course are:

1. To acquainted students with at least one operating systems software.
2. To provide students with skills in developing codes or algorithms that can manage selected system resources.
3. To provide students with scripting knowledge and skills, which can be used to measure the performance of selected components of a computer system.

Learning Outcomes

On completion of this course, the student shall be able to:

1. Explain basic concepts in at least one operating systems software.
2. Develop codes and algorithms that can manage selected system resources.
3. Demonstrate the ability to evaluate the performance of selected components of a computer system.

Course Outline

- | | |
|--|---------|
| 1. Introduction to the Operating System software | 6 hours |
| 2. Operating system Concepts and parameters | 6 hours |
| 3. File I/O | 6 hours |
| 4. System Processes | 6 hours |
| 5. Execution of Programs | 3 hours |
| 6. Threads | 6 hours |
| 7. Pipes and messages | 6 hours |

8. Memory Mappings 6
hours

9. Practicals 15hours

Mode of Delivery

Lectures and Practicals

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. The New KornShell Command And Programming Language, by Morris I. Bolsky, David G. Korn (Contributor).
2. Korn Shell Programming by Example, by Dennis O'Brien, David Pitts (Contributor).
3. The Korn Shell Linux and Unix Programming Manual (2nd Edn) by Anatole Olczak.
4. Linux Shell Scripting with Bash by Ken O. Burtch.
5. Unix Shell Programming by Stephen Kochan and Patrick Wood (third Edition).

INDUSTRIAL TRAINING II

Course Name	INDUSTRIAL TRAINING II
Course Level	YEAR 3 SEMESTER 2
Course Credit	4CU
Contact hours	60
Course code	SWE 3207

Course Description

Industrial Placement promotes and develops long-term strategic teaching and research alliances between higher education institutions and industry. It plays a crucial role in promoting and facilitating practical training by providing students the opportunity to apply advanced knowledge and skills acquired in class to real world industry based problems and also gain exposure to real life work experience.

Course Objectives

The course aims to enable students:

1. Understand the different types of Computing careers and their roles in a real world business organization.
2. Apply and translate computing knowledge and best practice to real industry based problems
3. Acquire soft skills to enhance effective participation in group based industry projects
4. Learn and practice good working ethics and communication skills with industry based staff and Supervisors.

Learning Outcomes

By the end of this course, the student should be able to:

- ✓ Explain the different types of Computing careers and their roles in a real world business organization.
- ✓ Demonstrate ability to apply and translate computing knowledge and best practice to industry based problems
- ✓ Demonstrate creativity and innovation in solving problems related to real-life projects
- ✓ Demonstrate soft skills for enhanced effective participation in group based industry projects

- ✓ Demonstrate and practice good working ethics, communicate effectively and exhibit good interpersonal skills with industry based staff and Supervisors.

Course Outline

The major areas that will be emphasized during industrial training include

- Orientation to Industry, Career Prospects & Human Resource Policy	4hours
- ICT Infrastructure in Industry	6hours
- Communication Network Infrastructure	4hours
- Information Management	10hours
- Systems Administration	4hours
- Website Development	4hours
- IT Quality Assurance	4hours
- Audit and Review ICT Infrastructure Maintenance Manuals	4hours
- Audit and Review IT and Communication Policies	4hours
- Audit and Review Information Security	4 hours
- Project Report Writting	12hours

Mode of Delivery

Mentorship

Course Assessment

• Evaluation Report by Industry Based Supervisor	40%
• Interview by Academic Supervisor	30%
• Written Industrial Training Project Report	30%

No particular reference will be used for this course unit, except whatever is recommended by the student supervisors.

YEAR FOUR SEMESTER ONE

COMPUTER AIDED DESIGN AND MANUFACTURING

Course Name: Computer Aided Design
Course Level: Year 4 semester 1
Course Credit: 4CU
Contact hours: 60 Hours

Course code: SWE 4101

Course Objective

Upon successful completion students will be able to: (i) Understand the basic concepts of CAD/CAM; (ii) Demonstrate knowledge and skill in applying computer and scientific principles related to solving engineering and manufacturing problems; and (iii) Develop an understanding of engineering design and learn how to provide technical assistance in managing CAD/CAM systems

Content:

Topics include:

Concepts of modelling in CAD/CAM; **5hrs**

Designing of specifications for interactive software packages in grafting; **5hrs**

Designing and manufacturing applications; **5hrs**

Design and development of algorithms; **5hrs**

Methodologies of problem solving in design; **5hrs**

Role of geometric models in CAD/CAM; **5hrs**

Techniques of displaying models using computer optimization; **5hrs**

Concepts of data representations in CAD/CAM problems; **5hrs**

Discussion on large numbers of case studies involving software development of various problems in CAD/CAM. **5hrs**

Modes of assessment

- Tests 20%
- Assignments 20%
- Examination 60%

References

1. Ibrahim Zeid, CAD/CAM, Tata McGraw Hill, New Delhi.
2. J. Rooney & P.Steadman, Principles of Computer Aided Design, Pitman/ Open University, London.
3. Joe Rooney & Philip Steadman, Computer Aided Design, Pitman/Open University, London.
4. Glen Mallineuse, Computational Concepts and Methods, Kogan Page Ltd. London.
5. Daniel L. Rayan, Computer Aided Graphical Design, Marcel Dekker, New York.

COMPUTER GRAPHICS ENGINEERING

COURSE NAME	GRAPHICS ENGINEERING
COURSE LEVEL	YEAR 4 semester 1
COURSE CREDIT	4CU
CONTACT HOURS	45 HOURS
Course code	SWE 4102

Course Description

This course introduces students to the basic principles of computer graphics. Students are exposed to the knowledge and skills of using computer graphics to portray real or imagined concepts.

Course Objectives

The course aims to:

1. Introduce students to computer graphics systems and concepts.
2. Provide students with an in-depth knowledge of the creation and analysis of computer graphics objects.

Learning Outcomes

Upon successful completion of this course students should be able to:

1. Explain computer graphics systems and concepts.
2. Demonstrate the ability to create and analyze computer graphics objects.

Course Content

The topics to be covered in this course include:

- | | |
|---|----------|
| 1. Computer graphics principles, terminologies and applications | 3 hours |
| 2. Graphics hardware | 3 hours |
| 3. Graphics design | 3 hours |
| 4. 2D and 3D transformations | 6 hours |
| 5. Graphics algorithms | 5 hours |
| 6. Simulating graphics | 4 hours |
| 7. Graphics visualisation | 3 hours |
| 8. Graphics standards | 3 hours |
| 9. Practical Lessons | 20 hours |

Mode of Delivery

Lectures, practicals, discussions, demonstrations

Course Assessment

- Tests 20%
- Assignments 20%
- Examination 60%

Reading materials

1. Computer Graphics (2nd Edition C Version) - Hearn, D., Baker, M.P. Prentice-Hall, 1997 (Entry Level book, good Illustrations): s.n.2176244-2nd ed. C version 1997, s.n. 2156999-2nd ed. 1994, s.n. 2033430-1986.
2. Computer Graphics - Principles and Practice (2nd Edition in C). J.D.Foley, A.Van-Dam, S.K.Feiner and J.F.Hughes. Addison-Wesley, 1996. (High level book, good in depth cover of many course aspects.): s.n.2186305-2nd ed. in C 1996, s.n.2092107-2nd ed.1990 .
3. Interactive Computer Graphics - Functional, Procedural and Device-Level Methods. P.Burger and D.Gillies. Addison-Wesley, 1989. (Little correspondence with course, old fashioned.) s.n.2096479.
4. Advanced Animation and Rendering Techniques. A.Watt and M.Watt. Addison-Wesley, 1992. (Extended material, for further reading and special interests.) s.n.2141145.
5. Programming Windows with MFC. Jeff Prosise, Microsoft Press; 2nd edition, 1999. (The book on MFC).
6. OpenGL Programming Guide. Woo, Manson., et al. Addison-Wesley, 1999 (official OpenGL redbook). s.n.2210662 -3rd ed. 1999, s.n. 2141236-1st ed. 1993.

FORMAL METHODS

Course Name
Course Level
Course Credit
Contact hours

FORMAL METHODS
Year 4 Semester I
3CU
45 Hours

Course code

SWE 4103

Brief Course Description

This course presents some formal notations that are commonly used for the description of computation and computing systems, for the specification of software and for mathematically rigorous arguments about program properties.

Course Objectives

The aims of this course are:

- 1 To introduce students to the mathematical notation and terminology used in formal specification
- 2 To equip students with knowledge of the fundamental principles in formal methods including laws and theorems
- 3 To equip students with an understanding of techniques and procedures required to solve practical problems
- 4 To impart programming skills by writing numerical programs required to solve the practical problems mentioned above

Learning Outcomes

Upon completion of this course students shall be able to:

- 1 Demonstrate knowledge including the mathematical notation and terminology of formal methods
- 2 Describe the fundamental principles including laws and theorems arising from formal methods concepts
- 3 Apply formal techniques and procedures to solve practical problems
- 4 Demonstrate programming skills by writing numerical programs, (e.g. Matlab) to solve practical problems

Course Outline

The topics to be covered in this course include:

1	Predicate Logic Specification	8 hours
2	Tools and systems of formal notations	9 hours
3	Algebraic Specification	10 hours
4	Other Topics e.g. Statecharts, Automatic program synthesis	8 hours
5	Tutorials (20 hours)	10 hours

Mode of Delivery

Lectures and tutorials

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

1. An Introduction to Formal Methods, by Antoni Diller, 2nd edition, Wiley, (June 1994), ISBN-10: 0471939730
2. Logic in Computer Science: Modelling and Reasoning about Systems, by Michael Huth and Mark Ryan, Cambridge University Press; 2nd Edition (August, 2004), ISBN-10: 052154310X
3. Formal Methods and Models for System Design: A System Level Perspective, by Gupta, R., Le Guernic, P. Shukla, S.K. and Talpin, J.P. (Eds.), 2004, ISBN: 978-1-4020-8051-7

SOFTWARE DESIGN PATTERNS

Course Name	SOFTWARE DESIGN PATTERNS
Course Level	Year 4 semester 1
Course Credit	3 CU
Contact hours	45
Course code	SWE 4104

Course Description

This course explores advanced principles of object-oriented design by studying key software design patterns. The patterns are drawn from a variety of sources and illustrated through examples and case studies. These are presented in an object oriented programming language e.g. Java, C++ or C sharp.

Course Objectives

The Course aims to provide students with understanding and skills in:

1. Designing effective and maintainable complex software systems of high quality by reusing/applying design patterns
2. Evaluating a design, identifying common problems, and how to fix these problems through refactoring

Learning Outcomes

By the end of this course, the student should be able to:

1. Describe the purpose of design patterns.
2. Describe a range of software patterns giving examples of their use
3. Understand the ways that design patterns are documented and classified.
4. Select and apply design patterns to practical software design problems
5. Select and apply design patterns to realistic business situations through a series of hands-on exercises

Course Outline

The major areas that will be covered include:

- | | |
|---|--------|
| 1. Introduction to design patterns | 3Hours |
| 2. Creational, structural and Behavioral Patterns | 6Hours |
| 3. Relationship between Patterns | 3Hours |

4. Patterns for concurrent and distributed systems	3Hours
5. Advanced Patterns	3Hours
6. Automation of Patterns and Anti-patterns	3Hours
7. Introduction to Aspect-Oriented design	3Hours
8. Introduction to business process modelling	3Hours
9. Patterns for coding, testing and concurrency control.	3hours
10. Practical and Case Studies of Applications of Design Patterns	15Hours

Mode of Delivery

Lectures, Tutorials, Practical's

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

Reading materials

1. Erich Gamma et al. Design Patterns
2. Wolfgang Pree. Design Patterns for Object-Oriented Software Development
3. Frank Buschmann, Kevlin Henney, Douglas C. Schmidt On Patterns and Pattern Languages

ARTIFICIAL INTELLIGENCE

Course title: Artificial Intelligence

Course level: Year 4 semester 1

Course credit: 3 CU

Course code: SWE 4105

DESCRIPTION

The course is designed to expose students to sophisticated computer programming in the development of smart systems. It covers Problems and Searches, Heuristic Search Techniques, Knowledge Representation, Game Playing, Planning, Understanding, Natural Language Processing, Parallel & Distributed AI, Learning, Connectionist Models, Common Sense, Expert Systems, Perception and Action.

EXPECTED LEARNING OUTCOMES

By the end of the course unit, students should be able to;

- Represent intelligent behaviour in terms of agent.
- Search a space of answers for a solution to a problem in practical time.
- Represent problems in terms of logic and deduction.
- Automate creation of complex plans in complex and unknown environments.
- Create Logical representations of uncertainty, and rational decision making in uncertain environments.
- Represent automated creation of new knowledge from examples and previous knowledge.

COURSE CONTENT

Introduction 9hrs

- Intelligent Agents – Agents and environments - Good behaviour – The nature of environments – structure of agents - Problem Solving - problem solving agents – example problems – searching for solutions – uniformed search strategies - avoiding repeated states – searching with partial information.

Searching Techniques 9hrs

- Informed search and exploration – Informed search strategies – heuristic function – local search algorithms and optimistic problems – local search in continuous spaces – online search agents and unknown environments - Constraint satisfaction problems (CSP) – Backtracking search and Local search for CSP – Structure of problems - Adversarial Search – Games – Optimal decisions in games – Alpha – Beta Pruning – imperfect real-time decision – games that include an element of chance.

Knowledge Representation 9hrs

- First order logic – representation revisited – Syntax and semantics for first order logic – Using first order logic – Knowledge engineering in first order logic - Inference in First order logic – prepositional versus first order logic – unification and lifting – forward chaining – backward chaining - Resolution - Knowledge representation - Ontological Engineering - Categories and objects – Actions - Simulation and events - Mental events and mental objects.

Learning 9hrs

Learning from observations - forms of learning - Inductive learning - Learning decision trees - Ensemble learning - Knowledge in learning – Logical formulation of learning – Explanation based learning – Learning using relevant information – Inductive logic programming - Statistical learning methods - Learning with complete data - Learning with hidden variable - EM algorithm - Instance based learning - Neural networks - Reinforcement learning – Passive reinforcement learning - Active reinforcement learning - Generalization in reinforcement learning.

Applications 9hrs

- Communication – Communication as action – Formal grammar for a fragment of English – Syntactic analysis – Augmented grammars – Semantic interpretation – Ambiguity and disambiguation – Discourse understanding – Grammar induction - Probabilistic language processing - Probabilistic language models – Information retrieval – Information Extraction – Machine translation.

MODE OF DELIVERY

18. Lectures

19. Reading assignments

20. In class discussions

21. Guided discovery

INSTRUCTIONAL MATERIALS/EQUIPMENT

- Whiteboard and Markers
- LCD Projectors
- Lab Exercises

COURSE ASSESSMENT

- | | |
|--|-----|
| • Continuous assessment tests | 20% |
| • Group and individual project (course work) | 20% |
| • End of semester examination | 60% |

Total	100%
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READING MATERIALS

Franklin, Stan (April 1997). *Artificial Minds* MIT Press; ISBN: 0262561093 ;

Gershenfeld, Neil A. (2000). *When Things Start to Think*. Owl Books; ISBN: 080505880X

Minsky, Marvin L (March 1988). *The Society of Mind*. Simon & Schuster; ISBN: 0671657135

Moravec, Hans P. (May 2000) .*Robot : Mere Machine to Transcendent Mind*. Oxford Univ Pr (Trade); ISBN: 0195136306.

SOFTWARE ENGINEERING INDUSTRIAL MINI PROJECT II

Course Name	SOFTWARE ENGINEERING INDUSTRIAL MINI PROJECT II
Course Level	Year 4 semester 1
Course Credit	4CU
Contact hours	60
Course code	SWE 4106

Course Description

The course introduces students to the integrated team approach to undertaking software Engineering projects from the point of requirements analysis to deploying a tested and well-engineered solution, with the assistance of a Project Supervisor (s). It also plays a crucial role in exposing students to building highly effective teams which is very important in “real life” industrial experience.

Course Objectives

The course aims to provide students with understanding and skills in:

1. Project management concepts and roles
2. Planning and controlling projects
3. Considerations for commercialization of software products
4. Software Requirements Analysis, specification and Design
5. Choosing appropriate Software Engineering Processes, Tools and Methods
6. Professional Software construction
7. Software maintenance and Management
8. Software documentation
9. Professional Software quality assurance and Testing

Learning Outcomes

By the end of this course, the student should be able to:

- ✓ Explain the nature of Software Engineering projects and their management
- ✓ Demonstrate understanding of Finance Planning, Budgeting and Costing in Software Engineering Projects
- ✓ Integrate the concepts of Risk Management, Human resource management, control of time and deadlines through application of tools and techniques

- ✓ Demonstrate the ability to develop a business plan as part of the Project Proposal for their choice of research project.
- ✓ Demonstrate creativity and innovation in the developing software solutions
- ✓ Develop the requirements and design specifications
- ✓ Attest and practice high-quality Project Management skills in enhancing individual effectiveness and productivity
- ✓ Develop Software using state of the art technologies
- ✓ Document the software in accordance to well known practices

Course Outline

The major areas that will be covered include:

– Project Proposal and Business Plans	6 Hours.
– Requirements Analysis, Specifications and Design	8 Hours.
– Project Management	4 Hours
– Software Methodology, Tools and Techniques	6 Hours
– Software Construction	20 Hours
– Software Quality Assurance and Testing	10 Hours
– Project Report Write Up	6 Hours

Mode of Delivery

Presentations, Project supervision

Course Assessment

• Project Proposal	10%
• Project Demonstration	50%
• Project Report	40%

No particular reference will be used for this course unit, except whatever is recommended by the student supervisors.

CLIENT SERVER PROGRAMING FOR APPLICATIONS

Course name: Client Server Programing for Applications

Course Level: year 4 semester 1

Credit units: 4

Contact hours: 60

Course code: SWE 4107

Course description

This course provides an introduction to constructing complete information systems based on the client-server model. On the client side, we introduce graphical user-interfaces, their design and implementation, as well as commonly used tools such as database access clients and report generators. On the server side we introduce database management systems and the use of server side programming tools that provide connectivity for clients and access to database systems. Along the way, students are introduced to the basics of distributed computing and computer networks.

Course Objective

This course will enable Students to:

- Construct a complete end to- end information system solution. This will often be their first opportunity to construct a non-trivial system of software.
- Integrate the database management and networking skills with client server programming.
- Create 3-tier applications.
- Perform basic website design
- Perform advanced client side programming
- Demonstrate the use of debugging techniques in client server applications.
- Describe the steps involved in the design, development and implementation of clientserver software application projects

- Write, test, debug and document client-server application development procedures.

Learning Outcomes

- Build application programs with multithreaded client side graphical user interfaces.
- Create a multithreaded server application.
- Understand basic networking concepts using sockets and remote method invocation Employ I/O Object serialization.
- Design a relational database in third normal form.
- Define, query and modify relational database tables using using SQL.
- Use the JDBC API to connect a Java application with relational databases.
- Proficiently utilize the latest Eclipse Interactive Development Environment with the most recent Java JDK for developing client server and standalone applications.

Course Outline

- 3 key elements of Client/Server Computing (**2 hours**)
- 3 elements that comprise tuning of a Client/Server system (**2 hours**)
- Business critical versus Mission critical applications (**2 hours**)
- Benefits of Client/Server computing (**2 hours**)
- Open Software Foundation – OSF(**3 hours**)
- Distributed Computer Environment – DCE (**3 hours**)
- Distributed Management Environment – DME (**3 hours**)
- Multiprocessing(**2 hours**)
- Multithreading(**2 hours**)

- Disk Mirroring and Disk Duplexing (*4 hours*)
- Data Striping (*3 hours*)
- Remote Procedure Calls – RPC(*8 hours*)
- ANSI/ISO 3 Schema Architecture(*8 hours*)
- 1,2, & 3 Tier Client Server Architectures(*5 hours*)
- Transaction Managers (*6 hours*)
- consistency anomalies (*5 hours*)

Mode of Delivery

Class lectures and notes are designed to achieve the course objectives. You should read the assigned chapters before class, complete assignments on time participate in class.

Course Assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

References

- Eck, D. J. (2009) Programming: Introduction to Programming Using Java.

CreateSpaceIndependent Publishing Platform.

- Moffat, A., (2003) Programming, Problem Solving, and Abstraction. Pearson
- Kenneth, C. L.(1993) Programming Languages: Principles and Practice, (2nd Ed) Thomson Brooks/Cole.

YEAR FOUR SEMESTER TWO

PROFESSIONAL AND ETHICAL ISSUES IN COMPUTING

Course Name	PROFESSIONAL AND ETHICAL ISSUES IN COMPUTING
Course Level	Year 4 semester 2
Course Credit	3 CU
Contact hours	45
Course code	SWE 4201

Course Description

The course provides awareness of the legal, social, ethical, professional and contemporary issues commensurate with the practice of a computing career.

Course Objectives

The Course aims to provide students with understanding and skills on:

- 1 Standards and professionalism in a Computing Career
- 2 Legal environment within which a computing professional operates
- 3 The professional societies that govern the behavior of computing professionals.
- 4 Professional responsibilities to client, employer and public

Learning Outcomes

By the end of this course, the student should be able to:

- 1 Appreciate the role of standards for a practicing computing professional
- 2 Practice a computing career with a due regard to legal, ethical and social issues

Course Outline

The major areas to be covered include:

- | | |
|---|---------|
| 1 The computing profession and standards | 3 hours |
| 2 Computing and society | 3 hours |
| 3 Ethics and the social context of computing | 6 hours |
| 4 Methods and tools for analysis of ethical arguments | 3 hours |
| 5 Risks and liabilities of safety-critical systems | 3 hours |
| 6 Intellectual property rights | 2 hours |
| 7 Privacy and civil liberties | 2 hours |
| 8 Social implications of the Internet | 3 hours |
| 9 Computer Misuse | 5 hours |

- | | | |
|----|-------------------------------------|----------|
| 10 | ICT legal framework | 5 hours |
| • | Tutorials and group work (20 hours) | 10 hours |

Mode of Delivery

Lectures and Tutorials

Course Assessment

- | | | |
|---|-------------|-----|
| • | Tests | 20% |
| • | Assignments | 20% |
| • | Examination | 60% |

Reading materials

- 1 Charles B. Fleddermann, Engineering Ethics, 1st edition Prentice Hall 1999. ISBN 13: 9780137842247
- 2 Kemper, John D.; Sanders, Billy R., Engineers and Their Profession, Oxford University Press, USA ISBN-13: 9780195120578
- 3 Moodley, Krisen, Engineering, Business and Professional Ethics, Elsevier Science & Technology 2007, ISBN-13: 9780750667418

SOFTWARE MAINTENANCE AND EVOLUTION

COURSE NAME	SOFTWARE MAINTENANCE AND EVOLUTION
COURSE LEVEL	YEAR 4 semester 2
COURSE CREDIT	3CU
CONTACT HOURS	45 HOURS

Course code **SWE 4202**

Course description

This course shows how Software evolution helps to implement and revalidate possible major changes to a system without the need to predict how user requirements will change. An existing large system is never complete and continues to evolve. As it evolves, the complexity of the system grows and Software evolution ensures continuous reliability and flexibility of the system.

Course Objectives

The aims of this course are to teach students:

1. The concepts and importance of software systems evolution
2. The software evolution process
3. Models and Laws of software systems evolution
4. The causes and impact of software systems evolution
5. The properties of software systems evolution
6. How selected software systems are analyzed based on properties of their evolution
7. How to interpret the implications of software systems evolution.

Learning Outcomes

Upon completion of the course, students shall be able to:

1. Explain the concepts of software systems evolution
2. Demonstrate the understanding of the causes and impact of software systems evolution
3. Explain the laws of software systems evolution
4. Describe the properties of software systems evolution
5. Analyze selected software systems to understand properties of their evolution
6. Interpret the implication of Evolution for software systems.

Course Content

To achieve the objectives of this course, the topics covered include:

1. Separate compilation

3hours

2. Design issues and common design patterns	3 hours
3. Verification and validation	3 hours
4. Integrating components	3 hours
5. Documentation	2 hours
6. Issues in object-oriented programming	3 hours
7. Parallelism	3 hours
8. event-centered programming	3 hours
9. software reuse	3 hours
10. The Models and Laws of Software Evolution	4 hours
11. The forward and Reverse engineering and reengineering tools	6 hours
12. Legacy system evolution	3 hours
13. Tutorials (12 hours)	6 hours

Mode of Delivery

Lectures and Tutorials

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

1. Software evolution: a software maintenance challenge, by lowell jay arthur, john wiley & sons, 1988, asin: 0471628719.
2. Modernizing legacy systems: software technologies, engineering processes, and business practices by robert c. sea cord, Daniel plakosh, and grace a. lewis, addison-wesley pub co; 1stedition, 2003, isbn: 0321118847.
3. Practical software maintenance: best practices for managing your software investment by Thomas m. pigoski, john Wiley & sons, 1st edition, 1996, isbn: 0471170011.
4. Designing maintainable software, by Dennis d. smith, springer-verlag, 1999, isbn 0387987835.

ENTREPRENEURSHIP

COURSE NAME:	ENTREPRENEURSHIP
COURSE CODE:	SWE4203
LEVEL:	YEAR THREE SEMESTER ONE
CREDIT UNITS:	3
CONTACT HOURS:	45

Course description

The course introduces the students to the basic concepts in entrepreneurship, identification of business opportunities, business evaluation and analysis. It provides students with the skills needed to effectively identify, organize, develop, and manage own business ventures. This course is based on creativity and professional development foundations that should orient a student to take adventure, a personal journey, and a seize opportunity for business start-up. The course gives students an opportunity to make creative adjustments to meet personal needs and increase self-drive to achieving success.

Course objectives

A student that undertakes this course should be able to:

- Analyze how entrepreneurship aids the economy and understand how entrepreneurship is essential to the vitality of any economy developed or developing.
- Identify, evaluate, select, and exploit business opportunities and resources
- Perform a self-evaluation to match their own characteristics with that of an entrepreneur
- Carry out feasibility and viability of an investment opportunity
- Analyze and exploit the Entrepreneurial Environment provided by the political, socioeconomic and technological conditions.
- Ensure start-up, survival, sustainability of an investment opportunity and identify their own personal entrepreneurial potential, ability, and competences.

LEARNING OUTCOME

On completion of this course unit, the students will be able to:

- Identify, analyse, compare and assess different entrepreneurial opportunities;
- Compute, develop and integrate key elements of the business plan, synthesize and generalize main parameters influencing the outcome of the new business start-up frame;
- Perform individual self-assessment recognizing and distinguishing main personal characteristics;
- Analyse and apply various idea generation and problem solving techniques;
- Discover and illustrate successful examples of new business establishment cases.

Course Outline

- The Nature and Importance of Entrepreneurship *(2 hours)*.
- The Entrepreneurial Mind-Set *(2 hours)*.
- Entrepreneurial Intentions and Corporate Entrepreneurship *(2 hours)*.
- International Entrepreneurship Opportunities *(2 hours)*.
- Creativity, the Business Idea, and Opportunity Analysis *(5 hours)*.
- Intellectual Property and Other Legal Issues for the Entrepreneur *(2 hours)*.
- The Business Plan: Creating and Starting the Ventures *(8 hours)*.
- The Marketing Plan *(8 hours)*.
- The Organizational Plan *(2 hours)*.
- The Financial Plan *(3 hours)*.
- Sources of Capital *(2 hours)*.
- Informal Risk Capital Venture Capital, and Going Public *(4 hours)*.
- Entrepreneurial Strategy: Generating and Exploiting New Entries *(3 hours)*.
- Strategies for Growth and Managing the Implications of Growth *(3 hours)*.
- Going Global .Accessing Resources for Growth from External Sources. Ending the Venture *(3 hours)*.
- The phenomenon of outsourcing and off-shoring software development; impacts on employment and on economics *(3 marks)*.

- Consequences of globalization for the computer science profession (*3 hours*)
- Differences in access to computing resources and the possible effects thereof
- Cost/benefit analysis of jobs with considerations to manufacturing, hardware, software, and engineering implications(*3 hours*).

Mode of Delivery

The teaching and learning approaches will combine use of case studies, Keynote lectures, student-led seminar presentations, site visits, and mini research

Course Assessment

- Course work 40%,
- Final Exam 60%

References

- OlomiDR (2009) African Entrepreneurship and Small Business Development; Otme Publishers Ltd.
- Doring JM, (2007) Entrepreneurship-Strategies and Resources (International Edition) , New Jersey, Pearson Publishers Inc. 1.
- Hisrich R. D., Peters M. P. (2002) Entrepreneurship. 5th ed., McGraw-Hill / Irwin.
- Kuratko D.F., Hodgetts R.M. (2007) Entrepreneurship. 7th ed., Thompson, South Western.

SOFTWARE SECURITY

Course Name

SOFTWARE SECURITY

Course Level

Year 4 semester 2

Course Credit

4CU

Contact hours

60 Hours

Course code

SWE 4204

Brief description

This unit aims to introduce secure software development issues from the design stage, through to implementation, testing and deployment. Topics studied include the secure software development life cycle, secure software design principles, threat evaluation models, secure coding and development practices, software security testing, deployment and maintenance. Students are provided with a range of practical exercises and tasks to reinforce their skills including: identification of security bugs in programs written in different programming languages, design, implementation, and testing of secure concurrent and networked applications and identification of vulnerabilities in networked and mobile/wireless applications.

Outcomes

At the completion of this unit, students should be able to:

1. model the possible vulnerabilities and threats for a given application system;
2. apply appropriate methods for the design and realisation of secure software;
3. Analyse and evaluate security properties of concurrent and networked applications.
4. Identify contemporary threats to information security
5. Apply contemporary solutions to the threats of information security using appropriate tools and technique
6. Propose solution of prevent, identify and mitigate the threats to information security

Course Outline

- | | |
|--|---------|
| 1. Data security concepts and examples | 3 hours |
| 2. Software security threats | 6 hours |

3. software security controls	6 hours
4. software security Tools and Techniques	8 hours
5. Information security Risk analysis	6 hours
6. Information security and social aspects	6 hours
7. Practicals (20 hours)	10 hours

Mode of Delivery

Lectures and practicals

Course Assessment

• Tests	20%
• Assignments	20%
• Examination	60%

Reading materials

1. Michael T. Goodrich and Roberto Tamassia, *Introduction to Computer Security*, Addison Wesley, 2011. ISBN: 0-32-151294-4
2. Handbook of Applied Cryptography, by Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone, ISBN: 0-8493-8523-7 October 1996.
3. Bruce Schneier, *Applied Cryptography*, John Wiley & Sons, 1996. ISBN 0-471-12845-7. On-line version.
4. William Stallings, *Network Security Essentials: Applications and Standards*, Prentice Hall, 4th edition, 2010. ISBN 0-13-610805-9.

NUMERIC ANALYSIS

Course Name	Numerical analysis
Course Level	Year 4 semester 2
Course Credit	4CU
Contact hours	60 Hours

Course code	SWE 4205
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Brief course description

Course Objectives Upon completion of the course, the student should be able to:

(i) Demonstrate factual knowledge including the mathematical notation and terminology used in the course; (ii) Describe the fundamental principles including the laws and theorems arising from the concepts covered in this course; (iii) Apply course material along with techniques and procedures covered in this course to solve practical problems; and (iv) Write numerical programs, such as Matlab programs, to solve the above problems.

Course Content

- Introduction to numeric analysis 4hrs
- Numerical linear algebra. 6hrs
- numerical solution of systems of non-linear equations, 8hrs
- approximations. 6hrs
- Fast Fourier-Transformation. 8hrs
- numerical integration. 8hrs
- difference equations and numerical solution of ordinary differential equations. 10 hrs
- Problem solving is an important part of the course. 14 hrs

Modes of assessment

- | | |
|---------------|-----|
| • Tests | 20% |
| • Assignments | 20% |
| • Examination | 60% |

References

- Numerical Analysis, by Burden, R.L. and Faires, D.J, Eighth Edition, Brooks/Cole Publishing Co., Pacific Grove, CA, 2001. ISBN: 0-534-38216-9
- Golub, Gene H. and Charles F. Van Loan (1986). Matrix Computations (third ed.). Johns Hopkins University Press. [ISBN 0-8018-5413-X](#).
- [Higham, Nicholas J.](#) (1996). Accuracy and Stability of Numerical Algorithms. Society for Industrial and Applied Mathematics. [ISBN 0-89871-355-2](#).
- [Hildebrand, F. B.](#) (1974). Introduction to Numerical Analysis (2nd ed.). McGraw-Hill. [ISBN 0-07-028761-9](#).
- [Leader, Jeffery J.](#) (2004). Numerical Analysis and Scientific Computation. Addison Wesley. [ISBN 0-201-73499-0](#).
- [Wilkinson, J.H.](#) (1965). The Algebraic Eigenvalue Problem (Clarendon Press).
- [Kahan, W.](#) (1972). "'A survey of error-analysis," in Info. Processing 71 (Proc. IFIP Congress 71 in Ljubljana), vol. 2, pp. 1214–39, North-Holland Publishing, Amsterdam". (examples of the importance of accurate arithmetic).
- [Trefethen, Lloyd N.](#) (2006). "[Numerical analysis](#)", 20 pages. In: Timothy Gowers and June Barrow-Green (editors), Princeton Companion of Mathematics, Princeton University Press.

FINAL YEAR PROJECT

Course Name	FINAL YEAR PROJECT
Course Level	Year 4 semester 2
Course Credit	4CU
Contact hours	60

Course code **SWE 4206**

Course Description

This course provides the student with the opportunity to independently undertake a research project from the point of requirements analysis to deploying a tested and well-engineered solution, with the assistance of a Project Supervisor(s). The student has to demonstrate a high level of professionalism and best practice in terms of documentation and software engineering standards.

Course Objectives

The Course aims to provide students with understanding and skills on:

1. Developing of a concept and plan for a commercially viable software project
2. Determining requirements and coming up design specification
3. Choosing appropriate Software Engineering Processes, Tools and Methods
4. Professional Software construction and deployment
5. Software maintenance and Management
6. Professional Software quality assurance and Testing
7. Software project documentation

Learning Outcomes

By the end of this course, the student should be able to:

1. Explain the nature of Software Engineering projects and their management
2. Demonstrate understanding of Finance Planning, Budgeting and Costing in Software Engineering Projects
3. Integrate the concepts of Risk Management, Human resource management, control of time and deadlines through application of tools and techniques
4. Demonstrate the ability to develop a business plan as part of the Project Proposal for their choice of research project.
5. Demonstrate creativity and innovation in the developing software solutions
6. Develop the requirements and design specifications

7. Attest and practice high-quality Project Management skills in enhancing individual effectiveness and productivity
8. Develop Software using state of the art technologies
9. Document the software in accordance to well known practices

Course Outline

The major areas that will be covered include:

- | | |
|---|----------|
| 1. Project Proposal and Business Plans | 6 Hours |
| 2. Requirements Analysis, Specifications and Design | 8 Hours |
| 3. Project Management | 4 Hours |
| 4. Software Methodology, Tools and Techniques | 6 Hours |
| 5. Software Construction | 20 Hours |
| 6. Software Quality Assurance and Testing | 10 Hours |
| 7. Project Report Write Up | 6 Hours |

Mode of Delivery

Presentations and Project supervision

Course Assessment

- Project Proposal
- Project Demonstration
- Project Report

Appendix 1: List of the Academic staff

NO.	INSTRUCTOR AND QUALIFICATION	Specialization	Awarding Institution	EMPLOYMENT STATUS
1.	Evarist Nabaasa (phd)	Networks and Programming	MUST, Hunan - China	Full time
2.	John Businge (phd)	Software Engineering	Eindhoven, Groningen - Netherlands	Full time
3.	Richard Ssembatya (phd)	Information Systems	UCT, Makerere	Full time
4.	Fred Kagwa (phd)	Security	MUST, UT-Netherlands	Full time
5.	Angella Musiimenta (phd)	ICT4D	Manchester, Leeds-UK	Full time
6.	Pius Ariho (phd)	Computing Mathematics	UNB-Canada, MUST	Full time
7.	Safari Yonasi (MSc)	Software Engineering and ICT4D	Kwazulu Natal University, Makerere	Full time
8.	Keneth Baguma (Msc)phd student	Information Technology	UMU	Full time
9.	Moreen Kabarungi (Msc)	ICT4D	MUST	Full time
10.	Ruth Nakato (MSc)phd student	Business Computing	Makerere	Full time
11.	Atuhaire Ambrose(Msc)phd student	Networks, Computer Designing	MUST	Full time
12.	Aggrey Obbo(Msc)phd student	Programming, Networks and Dbs	Makerere	Full time
13.	Mugonza Robert(Msc)phd student	Cloud Computing and Big Data	Leeds, MUST	Full time
14.	Jozephine Ayebare(Msc)	Trends in Computing	Chalmer University, UCU	Full time

15.	Gloria Munguci(Msc student)	ICT4D, Software Engineering	MUST, UCU	Full time
16.	Tonny Enguau(Msc student)	GIS	Makerere, MUST	Full time
17.	Richard Ntwari(Msc)phd student	Databases, and IoTs	UMU	Full time
18.	Aaron Atuhe (Msc)) phd student	Databases, Software Engineering	KIU, UCU	Full time
19.	Tezira Wanyana(Msc)phd student	Compilers and System Softwares	Chenai, MUST	Full time
20.	Irene Magara ((Msc)student)	Hardware Programming	Makerere, Kyambogo	Full time
21.	Francis Kamuganga (Msc)	Programming and Networks	Makerere, MUST	Full time
22.	Deborah Natumanya (Msc)phd student	GIS and ICT4D	MUST	Full time
23.	Rogers Mwavu (Msc)	Programming, Mobile Apps	MUST	Full time
24.	Richard Kimera(MSc)	Mobile App, Innovations and Dbs	Hangdong Global University, MUST	Full time
25.	Martin Ngobye	Agent Technology	Hunan, Makerere	Full Time

Note: No new staff recruitment will be required for this program.

Appendix 2: Proposed Budget in Uganda shillings.

Nature of the program		Degree	No. of students	Tuition/year/Student (ugsh)	total per year (ugsh)
Day-Private		BSc.SE	50	2,000,000	100,000,000 pa
Functional fees			50	552,000	27,600,000 pa
Identity card			50	33,000	16,500,000 pa
NCHE			50	20,000	1,000,000 Pa
Application fees			50	27,500	1,375,000 pa
total income per year					146,475,000 pa
CURRENT EXPENDITURE					
1. marketing and advertising		10,000,000			
2. Research projects, workshops, seminars		3,500,000			
3. Textbooks	25 each 80000	2,000,000			
4. Printing and stationery		2,500,000			
CAPITAL EXPEDITURE					
5. Computers 25	2,000,000	50,000,000			
Total Expenditure		68,000,000			
Contribution to university budget	78,475,000				